Please cite this article in press as: Bay and Palumbi, Multilocus Adaptation Associated with Heat Resistance in Reef-Building Corals, Current Biology (2014), http://dx.doi.org/10.1016/j.cub.2014.10.044

Current Biology 24, 1–5, December 15, 2014 ©2014 Elsevier Ltd All rights reserved http://dx.doi.org/10.1016/j.cub.2014.10.044

Report

Multilocus Adaptation Associated with Heat Resistance in Reef-Building Corals

Rachael A. Bay[1,*] and Stephen R. Palumbi[1]
[1]Department of Biology, Hopkins Marine Station of Stanford University, Pacific Grove, CA 93950, USA

Summary

The evolution of tolerance to future climate change depends on the standing stock of genetic variation for resistance to climate-related impacts [1, 2], but genes contributing to climate tolerance in wild populations are poorly described in number and effect. Physiology and gene expression patterns have shown that corals living in naturally high-temperature microclimates are more resistant to bleaching because of both acclimation and fixed effects, including adaptation [3]. To search for potential genetic correlates of these fixed effects, we genotyped 15,399 single nucleotide polymorphisms (SNPs) in 23 individual tabletop corals, *Acropora hyacinthus*, within a natural temperature mosaic in backreef lagoons on Ofu Island, American Samoa. Despite overall lack of population substructure, we identified 114 highly divergent SNPs as candidates for environmental selection, via multiple stringent outlier tests, and correlations with temperature. Corals from the warmest reef location had higher minor allele frequencies across these candidate SNPs, a pattern not seen for noncandidate loci. Furthermore, within backreef pools, colonies in the warmest microclimates had a higher number and frequency of alternative alleles at candidate loci. These data suggest mild selection for alternate alleles at many loci in these corals during high heat episodes and possible maintenance of extensive polymorphism through multilocus balancing selection in a heterogeneous environment. In this case, a natural population harbors a reservoir of alleles preadapted to high temperatures, suggesting potential for future evolutionary response to climate change.

Results

Thermal Environment

On Ofu Island in the National Park of American Samoa, a natural temperature mosaic over a small spatial scale provides an ideal natural laboratory for studying temperature adaptation. Two adjacent backreef pools differ in temperature regime: the highly variable (HV) pool experiences temperatures as high as 35°C and as low as 24.5°C [4], whereas the moderately variable (MV) pool ranges from 32°C–25°C (Figure 1A; Figure S1 available online). Corals in these pools bleach at temperatures of 33°C–35°C [5, 6] but tolerate cool temperatures of 24°C for over a week (R.A.B. and S.R.P., unpublished data). As a result, temperature variability in the HV pool appears to be stressful at the extreme upper ranges. We have previously shown that colonies of *Acropora hyacinthus* (sp. E in [7]) from the HV pool are more heat resistant in standardized tests conducted over 24 hr [3] or 72 hr [5].

We mapped temperatures experienced by 23 *A. hyacinthus* colonies over a 7 month period, February 2011 to August 2011. Colonies in the HV pool surpassed 31°C, the approximate expected bleaching threshold for American Samoa (http://www.coralreefwatch.noaa.gov), 0.47%–0.95% of the time (average = 0.67%), whereas individuals from the MV pool spent only 0%–0.20% of the time above 31°C (average = 0.09%) (Figure 1B). The amount of time spent above 31°C among individuals is also significantly more variable in the HV pool (F test p = 0.01). Similar differences occur between pools at 32°, 33°, and 34°C thresholds.

Colony SNP Differences and F_{ST} Outliers

To identify single nucleotide polymorphism (SNP) correlates of environment, we constructed and sequenced cDNA libraries from these same 23 *A. hyacinthus* colonies from the HV pool (n = 10) and MV pool (n = 13). Mapping reads to the annotated transcriptome from Barshis et al. [6] showed 107,693 high-quality SNPs, of which 15,399 could be confidently called in all individuals. F_{ST} calculations between the HV and MV pools showed higher F_{ST} values than expected across the 15,399 SNPs for all F_{ST} greater than 0.05 (95.5% randomizations had corrected Kolmogrov-Smirnov test $p \leq 0.05$, Figure S2). An F_{ST} outlier test showed 2,807 significant outliers (false discovery rate [FDR]-corrected $p \leq 0.05$), of which 1,445 had $F_{ST} > 0.05$. Because such tests likely yield many false positives [8], we also conducted a bootstrap resampling scheme. We conducted multiple runs of the simulation-based outlier analysis and discarded loci that did not appear in at least 70% of analyses, a cutoff commonly used for phylogenetic analyses [9]: a total of 251 SNPs passed this bootstrap filter.

Temperature and Colony Genetics

As a second way to sort loci for environmental selection, we filtered outlier loci for those at which individual genotypes correlated highly with individual temperature measurements. We used permutational ANOVA to compare genotype with fraction of time spent above 31°C for each individual, a metric that is consistent across multiple seasons ($p < 0.01$; Figures 3A and S1). Among all SNPs, this analysis yielded 1,006 loci associated with temperature (uncorrected $p \leq 0.05$). Of these, 114 were also F_{ST} outliers with ≥70% bootstrap score and $F_{ST} > 0.05$. Because these loci passed tests for outlier status, high F_{ST}, and genotype correlation with colony temperature, we used these 114 SNPs as candidates for environmental selection (Table S1).

For each SNP, we denoted the most frequent allele as the "reference" allele and the other allele as the "alternate" or "minor" allele. One unexpected pattern was that minor allele frequencies for candidate SNPs were generally higher in the HV pool, instead of having been evenly distributed among populations as were the noncandidate SNPs (Figure 2). For example, a homolog of the deformed epidermal autoregulatory factor (*DEAF1*) contains eight candidate SNPs. For seven of these SNPs, all individuals in the HV pool possessed at least one copy of the alternate allele (Figure 3B). Yet in the MV pool, the reference homozygote was the most common genotype. Across all candidate SNPs, each HV pool coral had alternate alleles at, on average, 57 of the 114 candidate loci, compared

*Correspondence: rbay@stanford.edu

Please cite this article in press as: Bay and Palumbi, Multilocus Adaptation Associated with Heat Resistance in Reef-Building Corals, Current Biology (2014), http://dx.doi.org/10.1016/j.cub.2014.10.044

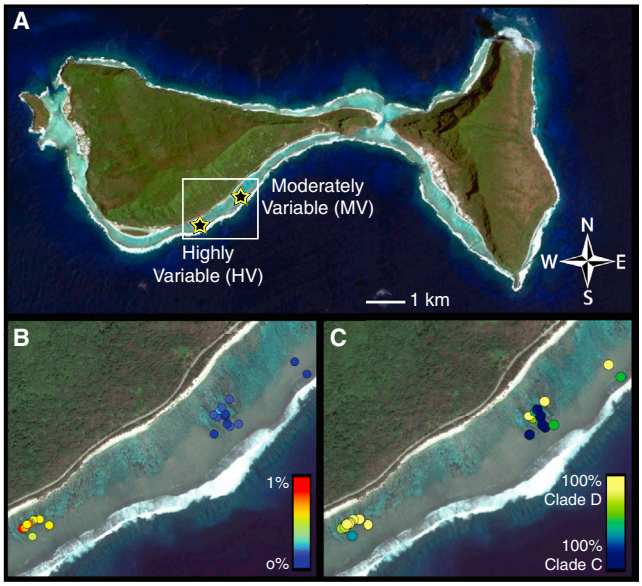


Figure 1. Sampling Sites, Thermal Environment, and Symbiont Type

(A) Map of sampled sites on Ofu Island, American Samoa. The white box highlights the region that is expanded in (B) and (C).

(B) Proportion of time spent above 31°C for each individual sampled from the highly variable (HV) and moderately variable (MV) pools.

(C) Results from symbiont typing analysis.

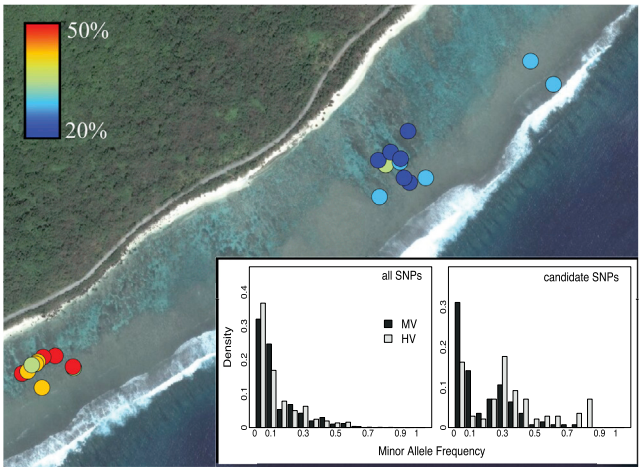


Figure 2. Polymorphism at 114 Candidate SNPs Is Maintained across a Heterogeneous Environment

Top: frequency of alternate (minor) alleles in candidate genes in the HV and MV pools. Colors indicate the percent of candidate loci with minor alleles for each individual. Inset shows histograms of minor allele frequency distributions. Left panel of inset shows distributions for all 15,399 SNPs, whereas the right panel shows distributions for 110 candidate SNPs for environmental selection.

to 32 loci in MV individuals ($p < 0.001$; Figure 2; Table S2). We also found a strong correlation between the fraction of time spent above 31°C for each colony and the fraction of loci with alternative alleles (Figure 3C: $R^2 = 0.74$, $p < 0.001$).

Because these differences might include a pool effect, not just a temperature difference, we also compared corals living in different temperature microclimates within pools. Although we had little power to detect significant trends within pools, we saw a positive trend within the MV when the number of alternate alleles at candidate loci was compared to thermal environment (Figure 3C). Although outliers in the HV pool obscured a similar trend, grouping colonies based on microclimate showed a similar pattern—individuals in warmer microclimates within the HV pool harbored more alternate alleles at candidate loci (Figure S3).

Overall colony genetic relationships also show higher variance in the HV pool. Our initial principal-component analysis (PCA) (Figure 4A) had four obvious outliers, three from the HV pool and one from the MV pool. The outlier from the MV pool (AH28) has the highest fraction of candidate loci with alternate alleles and lives in the third warmest microclimate. With these individuals removed, the PCA yielded six outliers, all from the HV pool. Individuals did not strictly cluster by pool; rather, warm microclimate colonies were scattered across the PCA, and HV individuals from the coolest microclimate group were grouped within the cooler MV cluster (Figure S4). Other principal-component (PC) axes were related more directly to temperature. PC4 was highly correlated with time above 31°C and was strongly influenced by variation in candidate SNPs: 34 of these loci (29.8%) were weighted in the top 5% on PC4.

Location, Temperature, and Symbionts

Every individual we sampled hosts a mixture of the thermally tolerant clade D of *Symbiodinium* and the less tolerant clade C [10–13]. However, most colonies in the HV pool were dominated (>85%) by clade D, whereas colonies from the MV pool more often hosted either clade C or D (Figure 1C). Because of this location effect, symbiont clade covaried with temperature across pools. However, within pools, the relationship was weak ($R^2 = 0.09$, $p = 0.16$), and there were no strong genetic correlations of any of our loci with symbiont type within the MV or HV pools. These data suggest that symbiont type does not play a major role at the loci we characterize here.

Gene Families Associated with Selection Candidates

Of 92 contigs that contain our 114 candidate SNPs, 68 had annotations from protein databases (Table S1) [6]. Candidate loci included proteins involved in heat stress response, including heat shock genes (3 contigs), protein recycling (3 contigs), and apoptosis. Although these gene families are known to be involved in heat stress reaction in corals [6, 14–16], formal gene ontology (GO) analysis showed no gene categories with significant enrichment (FDR $p > 0.05$) among our candidate genes.

Discussion

SNP variation at 114 loci is correlated with thermal environment in the coral *A. hyacinthus*. Many loci appear to be involved, rather than a few key genes. Similar to other studies of the evolution of complex environmental traits in natural populations, this result is consistent with a model of multigenic adaptation that is the cumulative result of a number of small-effect variants across many different cellular pathways [17, 18]. This pattern is similar to that observed in several forest tree species where adaptive variation is not the result of one allelic variant but rather the product of a number of small-effect variants [18]. In effect, thermal tolerance may be a quantitative trait conferred by alleles at many interacting loci. Such adaptive genetic variation over small spatial scales may provide a source of beneficial variation as climate change increases global ocean temperatures.

Please cite this article in press as: Bay and Palumbi, Multilocus Adaptation Associated with Heat Resistance in Reef-Building Corals, Current Biology (2014), http://dx.doi.org/10.1016/j.cub.2014.10.044

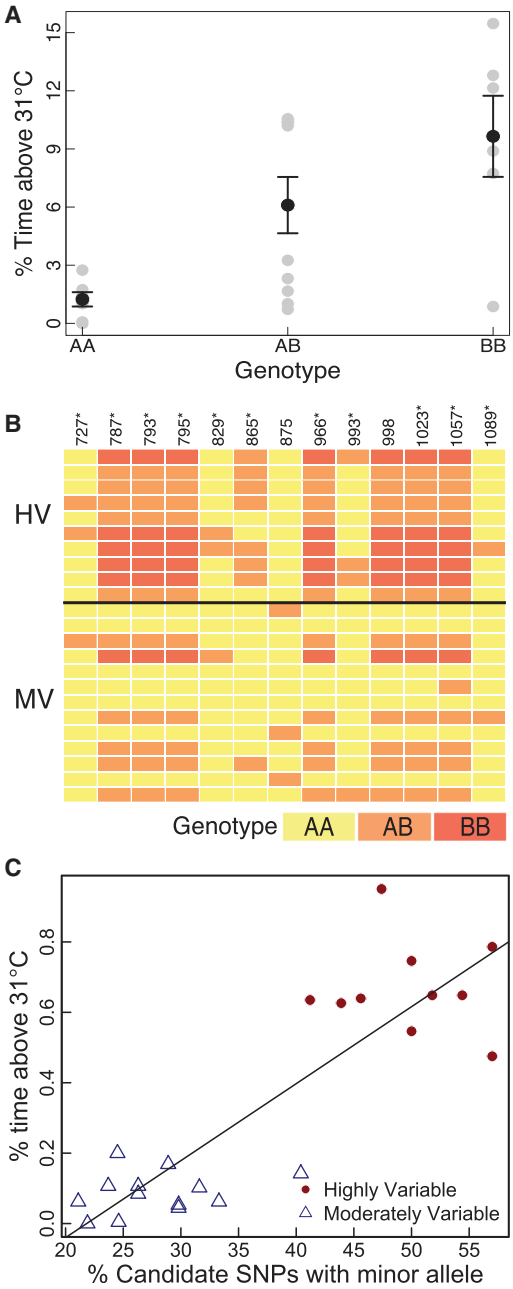


Figure 3. Relationship between Genotype and Thermal Environment

(A) Relationship between genotype and fraction of time spent above 31°C for the *DEAF1* homolog. Error bars indicate SEM for each genotype category.

(B) Genotypes at 13 SNPs of the *DEAF1* homolog, which could be genotyped in all individuals. Here, each row represents an individual coral colony, and each column represents an SNP. Column names indicate the position of the SNP within the contig from the *A. hyacinthus* transcriptome assembly [10]. Positions with amino acid changing SNPs are marked with an asterisk (*).

(C) Significant relationship ($R^2 = 0.74$, $p < 0.001$) between the number of loci at which a coral has alternative alleles for candidate loci and the temperature at which that coral lives.

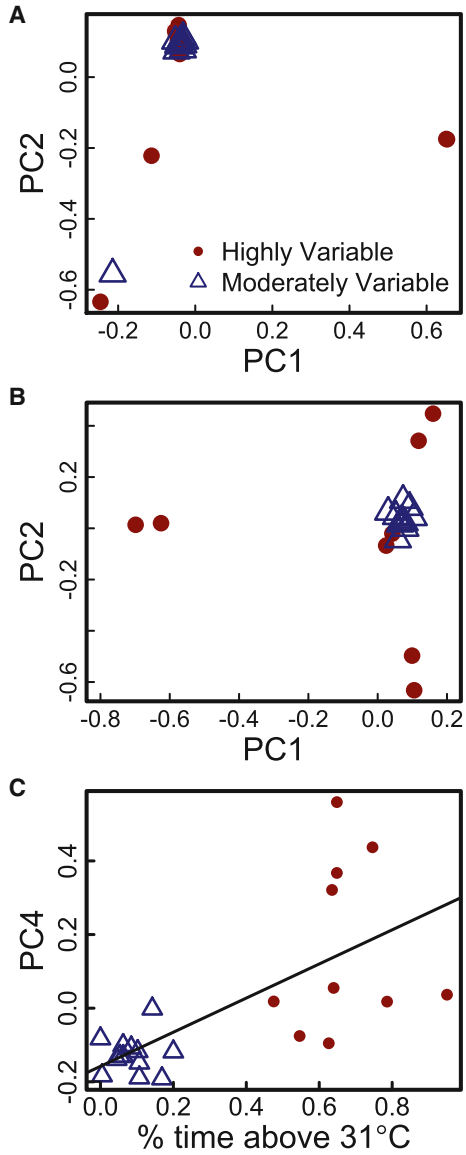


Figure 4. Principal-Component Analysis of 15,399 SNPs for *A. hyacinthus* Colonies on the Backreef of Ofu Island

Principal-component analysis (PCA) of 15,399 SNPs genotyped in 23 *A. hyacinthus* individuals.

(A) PCA with all individuals.

(B) PCA with four outlier individuals removed.

(C) Correlation ($R^2 = 0.42$, $p = 0.0004$) between the fourth principal component axis (PC4) and the proportion of time an individual spends above 31°C.

Patterns and Process of Natural Selection

Correlations between individual genetics and climate variables have been discovered in a number of species, including forest trees [19, 20], herbaceous plants [21, 22], and humans [23]. In our system, 114 SNPs, about 0.7% of the total, have signals of selection, a fraction similar to other marine studies that used comparable F_{ST} outlier approaches filtered by multiple criteria. For example, DeWit and Palumbi [24] found 3% of 21,579 SNPs to be F_{ST} outliers in populations of red abalone, and Hess et al. [25] found the same fraction in Pacific lamprey. However, those studies were done over much larger spatial scales (600–2,000 km). Corals such as *A. hyacinthus* have a 7–10 day period of free-swimming larval development [26]. Because pools are so close together compared to dispersal scale, selective differentiation at over 100 loci probably demands high selection pressure on settling coral larvae. The genetic load implied by this selection may require many settling coral larvae to die as a result of mismatches between genotype and environment [27].

Our data show that candidate SNPs for environmental selection have higher minor allele frequencies in the HV pool compared to the MV pool and that corals with a larger number of alternative alleles live at higher local temperatures (Figure 3A). These patterns are consistent with spatial balancing selection, when habitats with different selection pressures select for different alleles from a common gene pool [28]. Because the HV pool is small relative to the remainder of the reef, the overall frequencies of beneficial alleles are expected to be small. Colonies may be succeeding in several different ways depending on their genetic makeup, and it is possible that the number of loci at which an individual has the HV alternate allele relates to the individual's fitness (Figures 2 and 3A).

The observed pattern also suggests a fitness tradeoff: if an alternative allele is beneficial in the HV pool, but not detrimental elsewhere, then that allele should rise to fixation. The maintenance of polymorphism therefore suggests that either the two environments favor different genotypes or sufficient time has not passed for them to rise to fixation. Further studies on fitness effects of putatively adaptive variants will help differentiate between these possible scenarios.

Heat Tolerance and Local Microclimate

Corals in the HV and MV pools, including some of the same corals used here, show different levels of heat tolerance and also show marked differences in gene expression during experimental heat stress [6]. Palumbi et al. [3] performed reciprocal transplants between the HV and MV pools and showed that about half the increase of heat tolerance in the HV pool was due to acclimation of *A. hyacinthus*, and approximately half was due to fixed effects, including adaptation and epigenetic effects. These studies also examined differential gene expression during heat stress, and many of the dominant pathways are the same as those that we found to have candidate SNPs.

For example, studies of stress response in coral have found regulation of heat shock proteins and related transcription factors [6, 14, 15]. In our data, we see a signal for selection in proteins that interact with heat shock proteins, such as *HSC70*. The ubiquitin pathway is also known to be involved in heat stress and thermal tolerance [29, 30]. In addition, regulation of antioxidant defense and apoptosis genes, many of which were identified as selection candidates, is known to change expression in response to heat stress in corals [6, 15, 16]. Perhaps such physiological responses to stress have become incorporated into the genome through coding-region changes that alter gene function in an environment where individuals regularly experience stressful conditions. It is also possible that our focus on coding-region SNPs has missed numerous SNPs in gene regulatory regions, which appear to be common in genetic systems research (see [31] for review). Lastly, pool effects besides temperature may play a role in selection among our candidate loci.

Standing Variation and Climate Change Response

Our results pinpoint SNPs that are potentially adapted to different levels of thermal variability between pools. The data suggest standing variation in thermal tolerance within populations of *A. hyacinthus* and that selection in the HV pool produces genetic change at a number of loci. As ocean temperatures increase, beneficial genetic variants may become important for the maintenance of populations and species. Especially in highly dispersing species, these adapted populations might be able to seed more susceptible ones. Many current models that predict species response to climate change, however, assume that thermal tolerance is static not only within a population but also over time. For example, the model used by the National Oceanic and Atmospheric Administration to predict coral reef bleaching as well as long-term response assumes a fixed bleaching threshold that varies over latitude, but not temporally [32]. Studies have shown that incorporating adaptation into these models could greatly impact predictions for the future of coral reefs [33, 34].

Our results uncover a wealth of possible beneficial genetic variation and suggest a capacity for future adaptation of coral populations. Given that there is a role for genetic adaptation in thermal tolerance, future studies could focus on the rates at which this adaptation occurs. The relationship between the adaptive rate and the rate of temperature increase will determine whether natural selection can potentially mitigate negative impacts of anthropogenic climate change.

Supplemental Information

Supplemental Information includes Supplemental Experimental Procedures, four figures, and two tables and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.10.044.

Acknowledgments

We would like to thank Dan Barshis, Francois Seneca, Lupita Ruiz-Jones, and Dimitri Tolleter as well as Carlo Caruso and the U.S. National Park of American Samoa for help with sample collection; we would also like to thank the S.R.P. laboratory for many helpful discussions. This project was funded by the Gordon and Betty Moore Foundation.

Received: June 27, 2014
Revised: September 8, 2014
Accepted: October 14, 2014
Published: November 26, 2014

References

1. Etterson, J.R., and Shaw, R.G. (2001). Constraint to adaptive evolution in response to global warming. Science *294*, 151–154.
2. Hoffmann, A.A., and Sgrò, C.M. (2011). Climate change and evolutionary adaptation. Nature *470*, 479–485.
3. Palumbi, S.R., Barshis, D.J., Traylor-Knowles, N., and Bay, R.A. (2014). Mechanisms of reef coral resistance to future climate change. Science *344*, 895–898.
4. Craig, P., Birkeland, C., and Belliveau, S. (2001). High temperatures tolerated by a diverse assemblage of shallow-water corals in American Samoa. Coral Reefs *20*, 185–189.
5. Oliver, T.A., and Palumbi, S.R. (2011). Many corals host thermally resistant symbionts in high-temperature habitat. Coral Reefs *30*, 241–250.
6. Barshis, D.J., Ladner, J.T., Oliver, T.A., Seneca, F.O., Traylor-Knowles, N., and Palumbi, S.R. (2013). Genomic basis for coral resilience to climate change. Proc. Natl. Acad. Sci. USA *110*, 1387–1392.
7. Ladner, J.T., and Palumbi, S.R. (2012). Extensive sympatry, cryptic diversity and introgression throughout the geographic distribution of two coral species complexes. Mol. Ecol. *21*, 2224–2238.
8. Lotterhos, K.E., and Whitlock, M.C. (2014). Evaluation of demographic history and neutral parameterization on the performance of F_{ST} outlier tests. Mol. Ecol. *23*, 2178–2192.
9. Hillis, D.M., and Bull, J.J. (1993). An empirical test of bootstrapping as a method for assessing confidence in phylogenetic analysis. Syst. Biol. *42*, 182–192.
10. Baker, A.C., Starger, C.J., McClanahan, T.R., and Glynn, P.W. (2004). Coral reefs: corals' adaptive response to climate change. Nature *430*, 741.
11. Fabricius, K.E., Mieog, J.C., Colin, P.L., Idip, D., and van Oppen, M.J. (2004). Identity and diversity of coral endosymbionts (zooxanthellae) from three Palauan reefs with contrasting bleaching, temperature and shading histories. Mol. Ecol. *13*, 2445–2458.
12. Glynn, P.W. (1996). Coral reef bleaching: facts, hypotheses and implications. Glob. Change Biol. *2*, 495–509.

13. Rowan, R. (2004). Coral bleaching: thermal adaptation in reef coral symbionts. Nature *430*, 742.
14. Polato, N.R., Altman, N.S., and Baums, I.B. (2013). Variation in the transcriptional response of threatened coral larvae to elevated temperatures. Mol. Ecol. *22*, 1366–1382.
15. Császár, N.B.M., Seneca, F.O., and Van Oppen, M.J.H. (2009). Variation in antioxidant gene expression in the scleractinian coral *Acropora millepora* under laboratory thermal stress. Mar. Ecol. Prog. Ser. *392*, 93–102.
16. DeSalvo, M.K., Voolstra, C.R., Sunagawa, S., Schwarz, J.A., Stillman, J.H., Coffroth, M.A., Szmant, A.M., and Medina, M. (2008). Differential gene expression during thermal stress and bleaching in the Caribbean coral Montastraea faveolata. Mol. Ecol. *17*, 3952–3971.
17. Morgan, T.J., and Mackay, T.F.C. (2006). Quantitative trait loci for thermotolerance phenotypes in Drosophila melanogaster. Heredity *96*, 232–242.
18. Aitken, S.N., Yeaman, S., Holliday, J.A., Wang, T., and Curtis-McLane, S. (2008). Adaptation, migration or extirpation: climate change outcomes for tree populations. Evol. Appl. *1*, 95–111.
19. Eckert, A.J., van Heerwaarden, J., Wegrzyn, J.L., Nelson, C.D., Ross-Ibarra, J., González-Martínez, S.C., and Neale, D.B. (2010). Patterns of population structure and environmental associations to aridity across the range of loblolly pine (*Pinus taeda L.*, Pinaceae). Genetics *185*, 969–982.
20. Savolainen, O., Pyhäjärvi, T., and Knürr, T. (2007). Gene flow and local adaptation in trees. Annu. Rev. Ecol. Evol. Syst. *38*, 595–619.
21. Hancock, A.M., Brachi, B., Faure, N., Horton, M.W., Jarymowycz, L.B., Sperone, F.G., Toomajian, C., Roux, F., and Bergelson, J. (2011). Adaptation to climate across the *Arabidopsis thaliana* genome. Science *334*, 83–86.
22. Fournier-Level, A., Korte, A., Cooper, M.D., Nordborg, M., Schmitt, J., and Wilczek, A.M. (2011). A map of local adaptation in *Arabidopsis thaliana*. Science *334*, 86–89.
23. Hancock, A.M., Witonsky, D.B., Alkorta-Aranburu, G., Beall, C.M., Gebremedhin, A., Sukernik, R., Utermann, G., Pritchard, J.K., Coop, G., and Di Rienzo, A. (2011). Adaptations to climate-mediated selective pressures in humans. PLoS Genet. *7*, e1001375.
24. De Wit, P., and Palumbi, S.R. (2013). Transcriptome-wide polymorphisms of red abalone (*Haliotis rufescens*) reveal patterns of gene flow and local adaptation. Mol. Ecol. *22*, 2884–2897.
25. Hess, J.E., Campbell, N.R., Close, D.A., Docker, M.F., and Narum, S.R. (2013). Population genomics of Pacific lamprey: adaptive variation in a highly dispersive species. Mol. Ecol. *22*, 2898–2916.
26. Baird, A.H. (2001) The ecology of coral larvae: settlement patterns, habitat selection and the length of the larval phase. PhD thesis (Townsville: James Cook University).
27. Sanford, E., and Kelly, M.W. (2011). Local adaptation in marine invertebrates. Annu. Rev. Mar. Sci. *3*, 509–535.
28. Levene, H. (1953). Genetic equilibrium when more than one ecological niche is available. Am. Nat. *87*, 331–333.
29. Shahsavarani, H., Sugiyama, M., Kaneko, Y., Chuenchit, B., and Harashima, S. (2012). Superior thermotolerance of *Saccharomyces cerevisiae* for efficient bioethanol fermentation can be achieved by overexpression of RSP5 ubiquitin ligase. Biotechnol. Adv. *30*, 1289–1300.
30. Finley, D., Özkaynak, E., and Varshavsky, A. (1987). The yeast polyubiquitin gene is essential for resistance to high temperatures, starvation, and other stresses. Cell *48*, 1035–1046.
31. Torgerson, D.G., and Hernandez, R.D. (2012). Evolutionary signatures in non-coding DNA. In Rapidly Evolving Genes and Genetic Systems, R.S. Singh, J. Xu, and R.J. Kulathinal, eds. (Oxford: Oxford University Press), pp. 115–124.
32. Liu, G., Rauenzahn, J., Heron, S., Eakin, C., Skirving, W., Christensen, T., Strong, A., and Li, J. (2013). NOAA Coral reef watch 50 km satellite sea surface temperature-based decision support system for coral bleaching management. NOAA Technical Report NESDIS 143. March 2013. http://data.nodc.noaa.gov/coris/library/NOAA/CRCP/project/20370/NOAACRW50kmSatSSTemp-NESDISTechReport143.pdf.
33. Teneva, L., Karnauskas, M., Logan, C.A., Bianucci, L., Currie, J.C., and Kleypas, J.A. (2012). Predicting coral bleaching hotspots: the role of regional variability in thermal stress and potential adaptation rates. Coral Reefs *31*, 1–12.
34. Logan, C.A., Dunne, J.P., Eakin, C.M., and Donner, S.D. (2014). Incorporating adaptive responses into future projections of coral bleaching. Glob. Change Biol. *20*, 125–139.

24. J. A. Akins, S. N. Luo, P. D. Asimow, T. J. Ahrens, *Geophys. Res. Lett.* **31**, L14612 (2004).
25. J. A. Akins, T. J. Ahrens, *Geophys. Res. Lett.* **29**, 31-1–31-4 (2002).
26. Z. J. Liu *et al.*, *Solid State Commun.* **150**, 590–593 (2010).
27. D. Andrault *et al.*, *Nature* **487**, 354–357 (2012).
28. R. S. White, T. A. Minshull, M. J. Bickle, C. J. Robinson, *J. Petrol.* **42**, 1171–1196 (2001).
29. M. Murakami, Y. Ohishi, N. Hirao, K. Hirose, *Nature* **485**, 90–94 (2012).

ACKNOWLEDGMENTS

We thank M. Benbakkar, J. L. Devidal, A. Fabbrizio, G. Garbarino, P. Parisiadis, V. Svitlyk, and F. van Wyk de Vries for their help and three anonymous reviewers for fruitful comments. The MORB starting material was kindly provided by P. Schiano. This work is supported by Institut National des Sciences de l'Univers, l'European Synchrotron Radiation Facility, and the project Oxydeep of the Agence Nationale de la Recherche. This is Laboratory of Excellence ClerVolc contribution no. 99. All experimental data are presented in the supplementary materials.

SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/344/6186/892/suppl/DC1
Materials and Methods
Supplementary Text
Figs. S1 to S9
Tables S1 and S2
References (*30*–*34*)

6 January 2014; accepted 3 April 2014
10.1126/science.1250466

CORALS AND CLIMATE

Mechanisms of reef coral resistance to future climate change

Stephen R. Palumbi,* Daniel J. Barshis,† Nikki Traylor-Knowles, Rachael A. Bay

Reef corals are highly sensitive to heat, yet populations resistant to climate change have recently been identified. To determine the mechanisms of temperature tolerance, we reciprocally transplanted corals between reef sites experiencing distinct temperature regimes and tested subsequent physiological and gene expression profiles. Local acclimatization and fixed effects, such as adaptation, contributed about equally to heat tolerance and are reflected in patterns of gene expression. In less than 2 years, acclimatization achieves the same heat tolerance that we would expect from strong natural selection over many generations for these long-lived organisms. Our results show both short-term acclimatory and longer-term adaptive acquisition of climate resistance. Adding these adaptive abilities to ecosystem models is likely to slow predictions of demise for coral reef ecosystems.

Reef-building corals have experienced global declines resulting from bleaching events sparked by pulses of warm-water exposure (*1*–*4*). However, corals in naturally warm environments can have high resistance to bleaching temperatures and can survive heat exposure that would bleach conspecifics in cooler microclimates (*5*, *6*). Similarly, recent discovery of populations of acidification-resistant corals show that physiological or evolutionary mechanisms of environmental accommodation exist (*7*, *8*). Such populations are ideal test sites for research into the mechanisms of coral response to climate change.

Corals in adjacent backreef pools in the U.S. National Park of American Samoa on Ofu Island experience strong differences in temperature (*9*, *10*). In the highly variable (HV) pool, temperatures often exceed the local critical bleaching temperature of 30°C, reaching 35°C during strong noontime low tides (*6*). By contrast, the moderately bariable (MV) pool rarely experiences temperatures above 32°C. Corals in the HV Pool have higher growth rates (*9*, *10*), higher survivorship, and higher symbiont photosynthetic efficiency during experimental heat stress than conspecifics from the MV pool (*6*). These pools provide a powerful system to test the speed and extent of coral acclimatization and adaptation to warm-water conditions in the context of future climate change.

Department of Biology, Stanford University, Hopkins Marine Station, Pacific Grove, CA 93950, USA.
***Corresponding author. E-mail: spalumbi@stanford.edu** †Present address: Department of Biological Sciences, Old Dominion University, Norfolk, VA 23529, USA.

To test corals in their native habitats for physiological resistance to heat stress, we collected branches of the tabletop coral *Acropora hyacinthus* [cryptic species E (*11*)] and exposed them to experimental bleaching conditions. *A. hyacinthus* is a cosmopolitan species that constitutes a large percentage of hard coral cover on Pacific reefs and shows high levels of bleaching and mortality during large-scale bleaching events (*4*). We chose *A. hyacinthus* for this study because it is a dominant reef-builder and is especially sensitive to environmental stress, making its relative ability to acclimate or adapt extremely important to the future of coral reef ecosystems as climate change proceeds. We subjected branches of corals to a prescribed ramp in water temperature of 29° to 34°C for 3 hours, followed by an incubation for 3 hours at 34°C. These conditions mimic the natural increase in temperature observed in the HV pool during a tidal cycle. Experiments on fragments of tagged and monitored colonies showed that individuals native to the HV pool exhibit higher resistance to thermal stress, measured by retention of chlorophyll derived from photosynthetic symbionts, than corals from the MV pool (Fig. 1). The average retention of chlorophyll a after experimental heat stress was 80% in HV pool corals (Fig. 1C) but only 45% in MV pool corals (Fig. 1A, *t* test, $P < 0.00001$) compared with controls.

To test for acclimatization, we transplanted coral colonies of *A. hyacinthus* reciprocally from their native locations in the HV and MV pools to three transplant sites within each pool. We transplanted 6 colonies from the HV pool and 12 from the MV pool. After 12, 19, and 27 months, we tested transplanted colonies for thermal resistance. For 11 separate colonies, 22 of 23 paired bleaching

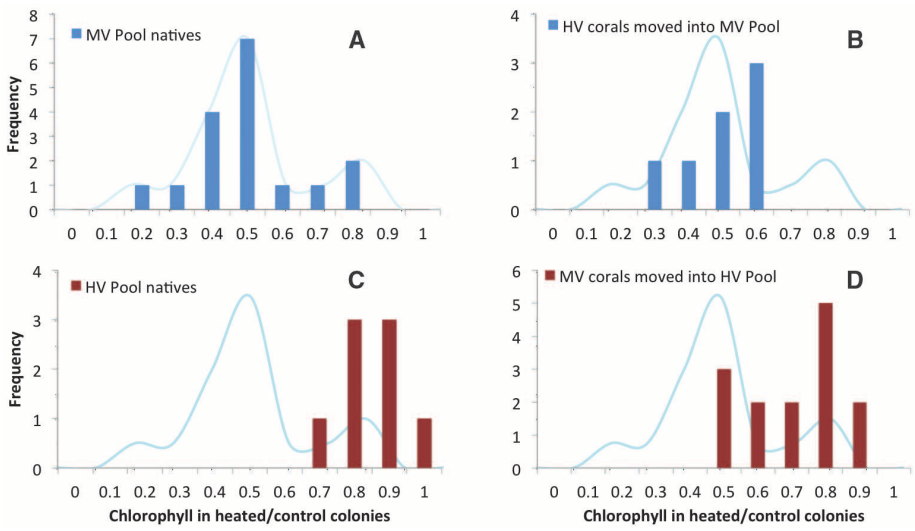


Fig. 1. Chlorophyll retention in coral colonies exposed to experimental heat stress compared with nonstressed controls. Upper panels show results from corals native to the moderately variable (MV) pool (**A**) and from corals moved into the MV pool (**B**). The lower panels are from corals native to the highly variable (HV) pool (**C**) and from corals moved into the HV pool (**D**). The smoothed curve reflects the distribution in (A) and is included in other panels for reference.

Downloaded from www.sciencemag.org on March 26, 2015

experiments show that corals acquired at least part of the heat sensitivity of the pool they were transplanted into. The experiments showed higher chlorophyll a retention during heat stress in colonies transplanted to the HV pool than in the same colony transplanted to the MV pool ($P < 0.0001$, paired t test, Fig. 2). Bleaching resistance did not vary with the time of transplant or season ($P > 0.80$).

Although the MV pool corals acquired heat resistance when moved to the HV pool, they did not achieve the resistance of native HV corals. MV pool corals transplanted to the HV pool retained less chlorophyll in bleaching experiments than corals native to the HV pool (compare Fig. 1, C and D; i.e., 67.5% and 80%, respectively; Student's t test, $P < 0.05$). By contrast, HV pool corals transplanted into the MV pool dropped their chlorophyll a retention to the same level as that of the MV pool natives (compare Fig. 1, A and B; i.e., 47% and 45%, respectively).

To understand the physiological changes associated with acquired heat resistance, we investigated gene expression in reciprocally transplanted corals. Six corals were sampled at noon on 30 August 2011, each from reciprocal transplant sites in the HV and MV pools. Reads from 12 mRNA extractions (one from each coral from each pool) were mapped to the *A. hyacinthus* assembly developed by Barshis *et al.* (*12*). Data pipelines and statistical procedures followed those of De Wit *et al.* (*13*). Transcriptome profiles of transplanted colonies show strong evidence for acclimatization. A two-way analysis of variance (ANOVA) of 16,728 coral genes detected 74 that changed significantly when comparing genetically identical coral fragments between the two pools [average 3.4-fold expression difference, false discovery rate (FDR)–corrected P value < 0.05, table S3). Among the 55 contigs with annotations, we found several transcription factors and cell signaling proteins (14 contigs), heat shock and chaperonin proteins (9 contigs), tumor necrosis factor receptor (TNFR)–associated factor (TRAF)–type proteins, cytochrome P450, and fluorochromes (Fig. 3A) that appeared to be involved in heat acclimation.

The ANOVA test also highlights 71 contigs with differential expression depending on the origin of colonies (two-way ANOVA experiment-wide FDR $P < 0.05$, table S2). These genes showed differences in expression levels depending strictly on the pool of origin, not on the final transplant site. In our previous work, members of the TNFR superfamily, which is involved in eukaryotic immune function and apoptosis (*14*, *15*), were constitutively up-regulated in native HV pool corals (*13*). In the current data set, one *TNFR* gene exhibited a 9.9-fold difference in expression in coral colonies native to the HV pool, whether they were living in the HV or MV pools (Fig. 3B), when compared with corals native to the MV pool. These results indicated that mediators of coral thermal resistance have fixed constitutive expression levels in either pool, perhaps representing signs of genetically based local adaptation.

Our transcriptome data also allowed us to test for changes in the proportions of symbiotic algae during coral acclimatization. *Symbiodinium* clades C and D are common in corals in American Samoa, and corals in warmer microclimates tend to have clade D symbionts (*16*). Though most colonies are dominated by one or the other of these clades, all colonies that we have tested host background populations of the other clade (*17*). To determine if coral host acclimation was accompanied by changes in symbiont proportion, we estimated the proportion of clade C and D by counting the transcriptome reads that mapped exclusively to small artificial test contigs of *ITS1*, *ITS2*, and the chloroplast *23S* gene, which distinguish clade C and D *Symbiodinium* (*17*). In our experiments, symbiont type explained a negligible fraction of the variation in bleaching resistance in common garden conditions (squared correlation coefficient R^2 = 0.15 and 0.06 in the HVP and the MVP transplants, respectively, $P > 0.30$, fig. S2). Our data also showed little shift in clade C versus clade D proportions as a result of transplantation. When clade C–dominated MV pool corals are moved to the HV pool, their proportion of clade D shifts from <1% to about 2%. Likewise, clade D–dominated HV corals moved to the MV pool maintain only about 4% clade C symbionts (Fig. 4). Additionally, there were no gene expression changes in *Symbiodinium* between transplants to the different pools, suggesting that little acclimation by symbionts occurred in the altered environmental conditions (*18*).

These experiments showed that some corals are capable of broad acclimatization to microclimate and developed enhanced resistance to bleaching without changing symbionts. We can place this capacity for acclimatization in an evolutionary context by comparing the natural phenotypic difference that we find between pools with the shift due to acclimatization after transplantation. Phenotypic change in evolutionary biology is often measured by the intensity parameter I, which is defined as the change in mean phenotype before versus after natural selection, divided by the standard deviation (SD) (*19*). In our case, we applied this concept to the spatial differences between pools instead of to temporal differences. The average phenotypic change from corals native to the HV pool versus the MV pool was 0.35 with a SD of 0.14 (compare Fig. 1, A and C). Thus, the phenotypic shift between pools was 2.5 SDs.

Here, I measures the difference in phenotype between coral populations subjected to different environments and is affected by phenotypic change caused by the fixed effects between the microclimates (denoted I_F), as well as by the acclimatization of individuals (denoted I_A), $I = I_F + I_A$. In this case, fixed effects include evolutionary adaptation, shifts in symbiont type, developmental changes, and epigenetics. Our transplant experiment allowed us to measure I_A as the average change in mean phenotype among acclimating individuals (0.214) divided by the SD (0.137), resulting in $I_A = 1.56$ (Fig. 1, A versus B and C versus D). This in turn allowed us to estimate I_F to be 0.94 SD units ($= I - I_A$).

These estimates indicated that these corals did acclimate to higher temperatures. The change in phenotype due to acclimatization (1.56 SD units) was similar to, but higher than, the change that we estimated is caused by fixed effects between pools (0.94 SD units).

We have provisionally extended this analysis to gene expression to illustrate the dual roles of acclimation and adaptation in gene expression shifts. There are 141 contigs for which corals native to the HV pool showed expression levels that differed by more than 1 SD from those of corals native to the MV pool and that were significant in the above two-way ANOVA analysis. Considering gene expression as a phenotype, these loci had an intensity $I > 1.0$. We also measured the contribution of acclimation to this phenotypic change by measuring gene expression in MV pool corals in the HV pool and vice versa. For these 141 contigs, the change caused by acclimation (I_A) averaged 42% of the phenotypic difference (fig. S4 and table S4), with the rest being caused by fixed effects (I_F). Loci with strong fixed effects included *TNFR* (as above) and a series of cellular transport loci with unknown function in corals. As in the

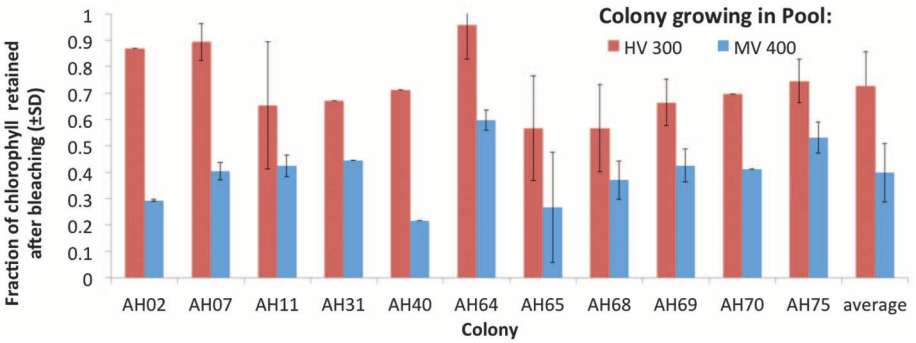


Fig. 2. Response of corals to reciprocal transplantation. The degree of bleaching experienced by a coral colony is measured by the ratio of chlorophyll that remains in experimentally heat-stressed colonies (N = 2 to 4 replicates per colony) compared to non–heat-stressed controls. Data are from 11 transplants into both the MV pool (blue) and HV pool (red). Error bars are SDs for colonies that were stressed-tested on two to three separate dates each.

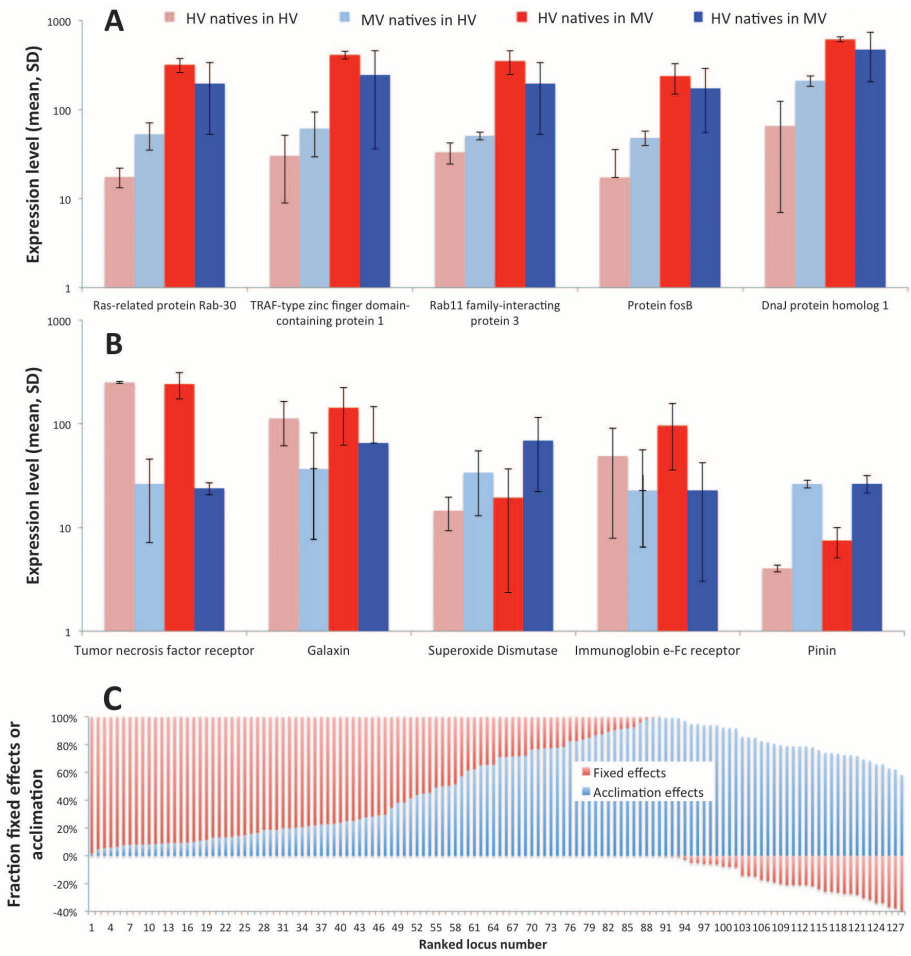


Fig. 3. Changes in gene expression between colonies on the basis of their transplant site (A) or original location (B). Light red represents corals native to the HV pool sampled after living in the HV pool. Light blue: MV natives sampled in MV pool. Dark red: HV natives in MV pool. Dark blue: MV natives in MV pool. Error bars are SDs. (**C**) The fraction of the gene expression difference between pools that is explained by fixed effects (red columns) versus acclimation (blue columns) for 128 loci that have large differences in expression between natives of the MV and HV pools. These 128 contigs correspond to the loci in table S4 and are listed by increasing level of acclimation between pools. Negative fixed effects occur when acclimation of transplanted corals exceeds the differences seen between natives. In our data, this occurs largely (33 of 35 cases) because transplanted HV pool corals in the MV pool have higher gene expression than MV pool natives.

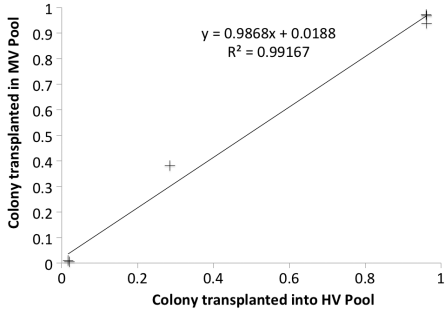


Fig. 4. High similarity of symbiont clade D proportion in six coral colonies transplanted to each of the HV pool and MV pool after 12 months of post-transplant growth.

case of bleaching phenotypes, fixed effects could be due to adaptive evolution, but could also be affected by developmental changes, epigenetics, and symbiont type. It is also possible there are residual effects that might eventually be erased by longer residence time. However, our experiments were conducted on branches that grew in situ after transplantation, and so our data are from tissues that did not experience the native pool environment. Loci with strong components of acclimation include *TRAF*, the signaling transducers for TNFR, as well as proteins encoded by *Ras* and *Rab*, transcription factors, and heat shock proteins.

The possibility that corals can acclimate to local differences has been suggested for decades: Conspecific corals at different latitudes show bleaching temperatures 1° to 2°C above local mean summer maximum sea-surface temperature (*20*, *21*), despite substantial differences in mean summer maximum temperatures. However, the mechanism generating this pattern has not been rigorously tested. Early analyses of the threat to corals from climate change (*1*, *2*) emphasized the potential for coral acclimatization or local adaptation to alter predictions of climate change effects. New models show that coral adaptation over a 40-year time frame could substantially change predictions for coral reef demise (*22*). But because future adaptation over many generations has been considered too slow for long-lived species such as corals, the role of individual acclimatization in coral environmental tolerance has been central to the debate on the future of reefs. The corals in our experiment achieved a larger bleaching phenotype shift (I_A = 1.56) due to acclimatization within 15 to 24 months than the shift due to fixed effects between habitats, a substantial acceleration in the rate of local matching of coral phenotype to thermal environment. Our results show that acclimatization can allow corals to acquire substantial high-temperature resistance more quickly than strong natural selection would produce.

We do not yet know how many coral species can acclimate or evolve. Several dozen coral species live and grow in the overheated backreef pools of Ofu (*23*), but whether all individual colonies have equal acclimatization ability, or if there is an upper thermal limit to acclimatization or adaptation (*24*), remain unknown. It is also probable that multiple stressors—from acidification and heat, for example—can reduce the ability of corals to respond (*25*). Thus, acclimatization alone cannot be expected to completely overcome the threat to corals from widespread bleaching events, especially if the onset of high-temperature stress is abrupt and sustained. In this regard, the tempo and severity of heat anomalies will be critical for effective coral acclimatization.

Persistence of populations through climate change demands biogeographic shifts of species (*26*–*28*), evolutionary adaptation of populations (*29*), or local acclimatization of individuals (*2*). For long-lived, sessile foundation species that create ecosystem habitat such as forest trees or reef-building corals, range shifts and evolution are predicted to be slow (*2*, *30*). Consequently, the rate and scope of acclimatization in these species to future environmental conditions is central to understanding the impact of climate change. For the fast-growing, shallow-water species that we studied, acclimatory and adaptive responses allowed them to inhabit reef areas with water temperatures far above their expected tolerances. How well other corals can similarly respond, and what the limits of these responses are, will determine how well current models accurately predict the future demise of coral reefs.

REFERENCES AND NOTES

1. O. Hoegh-Guldberg, *Mar. Freshw. Res.* **50**, 839 (1999).
2. T. P. Hughes *et al.*, *Science* **301**, 929–933 (2003).
3. T. P. Hughes, N. A. J. Graham, J. B. C. Jackson, P. J. Mumby, R. S. Steneck, *Trends Ecol. Evol.* **25**, 633–642 (2010).
4. K. E. Carpenter *et al.*, *Science* **321**, 560–563 (2008).
5. T. A. Oliver, S. R. Palumbi, *Mar. Ecol. Prog. Ser.* **378**, 93–103 (2009).
6. T. A. Oliver, S. R. Palumbi, *Coral Reefs* **30**, 429–440 (2011).

7. K. E. Shamberger et al., Geophys. Res. Lett. 41, 499–504 (2014).
8. K. E. Fabricius et al., Nat. Clim. Change 1, 165–169 (2011).
9. L. W. Smith, D. Barshis, C. Birkeland, Coral Reefs 26, 559–567 (2007).
10. L. W. Smith, H. H. Wirshing, A. C. Baker, C. Birkeland, Pac. Sci. 62, 57–69 (2008).
11. J. T. Ladner, S. R. Palumbi, Mol. Ecol. 21, 2224–2238 (2012).
12. D. J. Barshis et al., Proc. Natl. Acad. Sci. U.S.A. 110, 1387–1392 (2013).
13. P. De Wit et al., Mol. Ecol. Resour. 12, 1058–1067 (2012).
14. M. Karin, E. Gallagher, Immunol. Rev. 228, 225–240 (2009).
15. H. M. Shen, S. Pervaiz, FASEB J. 20, 1589–1598 (2006).
16. T. A. Oliver, S. R. Palumbi, Coral Reefs 30, 241–250 (2011).
17. J. T. Ladner, D. J. Barshis, S. R. Palumbi, BMC Evol. Biol. 12, 217 (2012).
18. D. J. Barshis, J. T. Ladner, T. A. Oliver, S. R. Palumbi, Mol. Biol. Evol. (2014).
19. J. Endler, Natural Selection in the Wild (Princeton Univ. Press, Princeton, NJ, 1986).
20. S. L. Coles, P. L. Jokiel, C. R. Lewis, Pac. Sci. 30, 155 (1976).
21. P. Jokiel, S. Coles, Coral Reefs 8, 155–162 (1990).
22. C. A. Logan, J. P. Dunne, C. M. Eakin, S. D. Donner, Glob. Change Biol. 20, 125–139 (2014).
23. P. Craig, C. Birkeland, S. Belliveau, Coral Reefs 20, 185–189 (2001).
24. J. H. Stillman, Science 301, 65 (2003).
25. J. E. Carilli, R. D. Norris, B. A. Black, S. M. Walsh, M. McField, PLOS ONE 4, e6324 (2009).
26. C. Parmesan et al., Nature 399, 579–583 (1999).
27. C. Parmesan, G. Yohe, Nature 421, 37–42 (2003).
28. T. L. Root, D. P. MacMynowski, M. D. Mastrandrea, S. H. Schneider, Proc. Natl. Acad. Sci. U.S.A. 102, 7465–7469 (2005).
29. C. Parmesan, Annu. Rev. Ecol. Evol. Syst. 37, 637–669 (2006).
30. O. Honnay et al., Ecol. Lett. 5, 525–530 (2002).

ACKNOWLEDGMENTS

The transcriptome data are archived at NCBI's Gene Expression Omnibus database (series record GSE56278: www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE56278). Funding was provided by The Gordon and Betty Moore Foundation through grant GBMF2629, the Schmidt Ocean Institute, and a fellowship from the NSF to N.T.K. We thank M. van Oppen, I. Baums, M. De Salvo, L. Bay, M. Morikawa, and O. Hoegh-Guldberg for comments on earlier versions of the manuscript. We also thank the U.S. National Park of American Samoa for permission to work on Ofu reefs and C. Caruso for logistical and research help.

SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/344/6186/895/suppl/DC1
Materials and Methods
Figs. S1 and S2
Tables S1 to S4
References (*31*–*35*)

27 January 2014; accepted 1 April 2014
Published online 24 April 2014;
10.1126/science.1251336

EVOLUTION

Ancient DNA reveals elephant birds and kiwi are sister taxa and clarifies ratite bird evolution

Kieren J. Mitchell,[1] Bastien Llamas,[1] Julien Soubrier,[1] Nicolas J. Rawlence,[1]* Trevor H. Worthy,[2] Jamie Wood,[3] Michael S. Y. Lee,[1,4] Alan Cooper[1]†

The evolution of the ratite birds has been widely attributed to vicariant speciation, driven by the Cretaceous breakup of the supercontinent Gondwana. The early isolation of Africa and Madagascar implies that the ostrich and extinct Madagascan elephant birds (Aepyornithidae) should be the oldest ratite lineages. We sequenced the mitochondrial genomes of two elephant birds and performed phylogenetic analyses, which revealed that these birds are the closest relatives of the New Zealand kiwi and are distant from the basal ratite lineage of ostriches. This unexpected result strongly contradicts continental vicariance and instead supports flighted dispersal in all major ratite lineages. We suggest that convergence toward gigantism and flightlessness was facilitated by early Tertiary expansion into the diurnal herbivory niche after the extinction of the dinosaurs.

Despite extensive studies, the evolutionary history of the giant flightless ratite birds of the Southern Hemisphere landmasses and the related flighted tinamous of South America has remained a major unresolved question. The ratites and tinamous, termed "palaeognaths" due to their shared basal palate structure, form the sister taxon to all other living birds (neognaths). The living ratites are one of the few bird groups composed largely of giant terrestrial herbivores and include: the emu and cassowary in Australia and New Guinea, the kiwi in New Zealand, the ostrich in Africa, and the rhea in South America. In addition, two recently extinct groups included the largest birds known: the moa from New Zealand (height up to 2 to 3 m, 250 kg in weight) (*1*) and elephant birds from Madagascar (2 to 3 m in height, up to 275 kg in weight) (*2*, *3*). Ratites have been believed to have originated through vicariant speciation driven by the continental breakup of the supercontinent Gondwana on the basis of congruence between the sequence of continental rifting and the presumed order of lineage divergence and distribution of ratites (*4*, *5*).

New Zealand is the only landmass to have supported two major ratite lineages: the giant herbivorous moa and the chicken-sized, nocturnal, omnivorous kiwi. Morphological phylogenetic analyses initially suggested that these two groups were each other's closest relatives (*6*, *7*), presumably diverging after the isolation of an ancestral form following the separation of New Zealand and Australia in the late Cretaceous ~80 to 60 million years ago (Ma) (*8*). However, subsequent studies suggest that kiwi are more closely related to the Australasian emu and cassowaries (*9*, *10*), whereas the closest living relatives of the giant moa are the flighted South American tinamous (*11*–*14*). The latter relationship was completely unexpected on morphological grounds and suggests a more complex evolutionary history than predicted by a model of strict vicariant speciation. By rendering ratites paraphyletic, the relationship between the moa and tinamous also strongly suggests that gigantism and flightlessness have evolved multiple times among palaeognaths (*12*, *13*).

Perhaps the most enigmatic of the modern palaeognaths are the recently extinct giant Madagascan elephant birds. Africa and Madagascar were the first continental fragments to rift from the supercontinent Gondwana, separating from the other continents (and each other) completely during the Early Cretaceous (~130 to 100 Ma) (*15*). Consequently, the continental vicariance model predicts that elephant birds and ostriches should be the basal palaeognath lineages (*16*). Most molecular analyses recover the ostrich in a basal position, consistent with a vicariant model. However, the phylogenetic position of the elephant birds remains unresolved, as cladistic studies of ratite morphology are sensitive to character choice and may be confounded by convergence (*17*), whereas DNA studies have been hampered by the generally poor molecular preservation of elephant bird remains (*18*).

We used hybridization enrichment with in-solution RNA arrays of palaeognath mitochondrial genome sequences and high-throughput sequencing to sequence near-complete mitochondrial genomes from both elephant bird genera: *Aepyornis* and *Mullerornis*. Phylogenetic analyses placed the two taxa, *Aepyornis hildebrandti* (15,547 base pairs) and *Mullerornis agilis* (15,731 base pairs), unequivocally as the sister taxa to the kiwi (Fig. 1 and fig. S1). This result was consistently retrieved, regardless of phylogenetic method or taxon sampling, and was strongly supported by topological tests (*19*). To our knowledge, no previous study has suggested this relationship, probably because of the disparate morphology, ecology, and distribution of the two groups. Elephant birds were herbivorous, almost certainly diurnal, and among the largest birds

[1]Australian Centre for Ancient DNA, School of Earth and Environmental Sciences, University of Adelaide, North Terrace Campus, South Australia 5005, Australia. [2]School of Biological Sciences, Flinders University, South Australia 5001, Australia. [3]Landcare Research, Post Office Box 40, Lincoln 7640, New Zealand. [4]South Australian Museum, North Terrace, South Australia 5000, Australia.
*Present address: Allan Wilson Centre for Molecular Ecology and Evolution, Department of Zoology, University of Otago, Dunedin, New Zealand. **†Corresponding author. E-mail: alan.cooper@adelaide.edu.au**

Synthesis

Predicting Responses to Contemporary Environmental Change Using Evolutionary Response Architectures

Rachael A. Bay,[1,*,†] Noah Rose,[2,*] Rowan Barrett,[3] Louis Bernatchez,[4] Cameron K. Ghalambor,[5] Jesse R. Lasky,[6] Rachel B. Brem,[7] Stephen R. Palumbi,[8] and Peter Ralph[9]

1. Center for Tropical Research, Institute for the Environment and Sustainability, University of California, Los Angeles, California 90095; 2. Department of Ecology and Evolutionary Biology, Princeton University, Princeton, New Jersey 08544; 3. Redpath Museum and Department of Biology, McGill University, Montreal, Quebec H3A 1B1, Canada; 4. Department of Biology, Institut de Biologie Intégrative et des Systèmes, 1030 Avenue de la Medecine Université Laval, Quebec City, Quebec G1V 0A6, Canada; 5. Department of Biology, Colorado State University, Fort Collins, Colorado 80523; 6. Department of Biology, Pennsylvania State University, University Park, Pennsylvania 16802; 7. Buck Institute for Research on Aging, Novato, California 94945; and Department of Plant and Microbial Biology, University of California, Berkeley, California 94720; 8. Hopkins Marine Station, Stanford University, Pacific Grove, California 93950; 9. Departments of Mathematics and Biology, University of Oregon, Eugene, Oregon 97403

Submitted May 17, 2016; Accepted November 18, 2016; Electronically published March 17, 2017

abstract: Rapid environmental change currently presents a major threat to global biodiversity and ecosystem functions, and understanding impacts on individual populations is critical to creating reliable predictions and mitigation plans. One emerging tool for this goal is high-throughput sequencing technology, which can now be used to scan the genome for signs of environmental selection in any species and any system. This explosion of data provides a powerful new window into the molecular mechanisms of adaptation, and although there has been some success in using genomic data to predict responses to selection in fields such as agriculture, thus far genomic data are rarely integrated into predictive frameworks of future adaptation in natural populations. Here, we review both theoretical and empirical studies of adaptation to rapid environmental change, focusing on areas where genomic data are poised to contribute to our ability to estimate species and population persistence and adaptation. We advocate for the need to study and model evolutionary response architectures, which integrate spatial information, fitness estimates, and plasticity with genetic architecture. Understanding how these factors contribute to adaptive responses is essential in efforts to predict the responses of species and ecosystems to future environmental change.

* These authors contributed equally to this work.
† Corresponding author; e-mail: rachaelbay@gmail.com.
ORCIDs: Rose, http://orcid.org/0000-0001-7129-4753; Ghalambor, http://orcid.org/0000-0003-2515-4981; Lasky, http://orcid.org/0000-0001-7688-5296; Brem, http://orcid.org/0000-0002-8587-9434; Palumbi, http://orcid.org/0000-0001-5195-1947; Ralph, http://orcid.org/0000-0002-9459-6866.

Am. Nat. 2017. Vol. 189, pp. 463–473. © 2017 by The University of Chicago. 0003-0147/2017/18905-56995$15.00. All rights reserved. This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0), which permits non-commercial reuse of the work with attribution. For commercial use, contact journalpermissions@press.uchicago.edu.
DOI: 10.1086/691233

Keywords: population genomics, environmental change, adaptation, plasticity, evolutionary prediction.

Introduction

As contemporary environments change at an unprecedented rate, species persistence will increasingly depend on the ability to respond and adapt to novel environmental conditions. Models used currently to predict species responses to environmental change often involve a climate envelope approach, assuming static environmental tolerance over time, such that organisms will track current preferred environmental conditions in the future (Fernandes et al. 2013; Pinsky et al. 2013). However, decades of empirical research contradict this assumption, as evolutionary and physiological studies have revealed that tolerance to rapid environmental change varies among individuals, among populations, and across species (Somero 2010; Hoffmann and Sgrò 2011) and has the potential to rapidly evolve (Hendry and Kinnison 1999; Reznick and Ghalambor 2001). The question of whether the rate of adaptation can keep up with unprecedented rates of human-induced environmental change—and whether this ability varies among species, populations, and ecosystems—remains an essential and unresolved question in evolutionary biology (Merilä and Hendry 2014).

Two major lines of inquiry have developed in response to this question. First, theoretical models of population genetic responses to environmental change have been created to explore the conditions under which populations can

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

evolve quickly enough to avoid extinction (Gomulkiewicz and Holt 1995; Hermisson and Pennings 2005; Chevin et al. 2010; Orr and Unckless 2014; Uecker et al. 2014). These models have clarified the parameter space in which rapid adaptation is expected to play an important role in population persistence and have demonstrated that these processes are expected to contribute significantly to future ecosystem function. Second, empirical studies have repeatedly identified a genetic basis for variation in environmental tolerance within populations, clarifying the architecture of local adaptation between populations and providing evidence of recent rapid adaptation and speciation across environmental gradients. These include common garden and experimental evolution studies examining local adaptation and evolution of environmental tolerance (Bradshaw and Holzapfel 2001; Collins and Bell 2004; Van Doorslaer et al. 2007; Paul et al. 2011), genetic monitoring (e.g., Umina et al. 2005; Balanyá et al. 2006), and more recently, analysis based on genome-wide sequence variation and transcriptomic profiling (Hancock et al. 2011; Pespeni et al. 2013; Bay and Palumbi 2014; Dixon et al. 2015; Ghalambor et al. 2015). These experiments show that contemporary adaptation and acclimation are playing a major role in the persistence of real-world populations.

These theoretical and empirical studies highlight the information we need to understand and predict future response to environmental change, which we call the evolutionary response architecture (fig. 1). This outlines types of data needed to answer outstanding questions related to predicting evolutionary response to environmental change such as the following: How can the direction, rate, magnitude, and limits of potential evolutionary change be inferred using genomic data from natural populations, and do estimates match observed outcomes? How do parameters like the amount of standing sequence variation, gene flow, population size, and genomic architecture vary among populations? Under what conditions will populations respond to environmental change through phenotypic plasticity, evolution, or both? In this synthesis, we review the key theoretical determinants of evolutionary responses to rapid environmental change and outline current methods that use genomic data to estimate these factors. Finally, we examine major challenges and knowledge gaps and discuss strategies for filling these gaps.

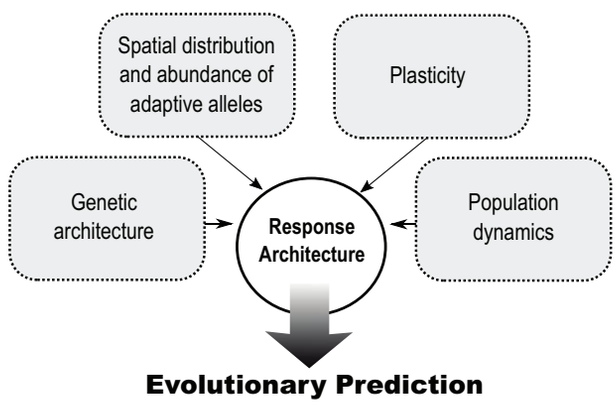


Figure 1: Conceptual map of response architecture for predicting evolution in response to environmental change.

Determinants of Population Responses to Environmental Change

A growing body of theoretical work examines conditions under which rapid environmental change is expected to drive rapid evolutionary change. These conditions include that there must be sufficient standing genetic variation in the relevant trait(s) and that adaptation must proceed quickly enough to avoid extinction (Orr et al. 2008; Messer and Petrov 2013; Carlson et al. 2014; Orr and Unckless 2014). This evolutionary response is further modified by several attributes of the system of interest. Standing genetic variation in environmental response may be highly polygenic or dependent on only a few loci (Lynch and Lande 1993; Burger and Lynch 1995; Gomulkiewicz and Holt 1995). This variation may be present within populations or stratified across populations or parts of a species' range (Ralph and Coop 2010; Uecker et al. 2014). Selection may vary due to spatiotemporal environmental gradients or frequency-dependent selection (Holt et al. 2004; Ralph and Coop 2015). Finally, plasticity may either promote or inhibit selection on environmental tolerance traits (Chevin et al. 2010). Theoretical studies have covered many portions of this range of possibilities, and a key step in applying theory will be to determine which sorts of models to use.

Genetic Architecture and Population Dynamics

Most studies have investigated the fixation dynamics of a trait determined by a single locus that is sufficient to rescue an isolated population exposed to an instantaneous environmental shift. The most fundamental insight from these studies is that, given an environmental tolerance allele with large benefit and high frequency in the starting population, fixation is likely and extinction can be averted (Hermisson and Pennings 2005; Orr and Unckless 2014). By contrast, if the selective advantage of the environmental tolerance allele is too low or if it is present in too few copies, it is very likely to be lost due to random genetic drift before it can contribute to adaptation (Hermisson and Pennings 2005; Barrett and Schluter 2008). Because reduced absolute fitness caused by rapid environmental change will in many cases reduce population sizes, populations may go extinct before adaptation can restore a positive population growth rate (Lynch and Lande 1993; Orr and Unckless 2014). The special case

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

in which environmental change reduces population mean absolute fitness to below replacement but extinction is averted by rapid evolutionary change is referred to as evolutionary rescue. However, these effects are strongly dependent on the maximum population growth rate, because a sufficient reproductive excess may be able to accommodate high levels of mortality without a decrease in adult population size if growth rate is not reduced to below replacement (Crow 1970).

Orr and Unckless (2008), for instance, combined these contributions to find the probability of adaptation from mutations or rare standing variation in a randomly mating population. If the population is declining at rate r (absolute fitness $1 - r$), then the total number of individuals that will ever live, if adaptation does not occur, is N_0/r, where N_0 is the initial population size; and if u is the per-generation mutation rate of the rescuing allele, there are expected to be uN_0/r such mutants, added to the initial number N_0p_0 present as standing variation. Each new mutant has probability $2s/v$ of establishing, so the expected number to escape drift is $2N_0s(u/r + p_0)/v$, and by Poisson approximation the probability that at least one new mutant establishes is

$$P_{\text{rescue}} = e^{-2N_0s(u/r+p_0)/v} \tag{1}$$

(Orr and Unckless 2008). Note that in this equation, $1 + s$ is the population growth rate of the new allele when rare (this is $s - r$ in Orr and Unckless 2008) and that N_0/v is the effective population size. Studies of standing variation for traits promoting environmental tolerance should therefore give strong insight into the likely evolutionary dynamics of a focal population in response to contemporary environmental change by providing estimates of starting allele frequencies (p_0) and selective coefficients (s) of environmental tolerance loci. However, understanding which genotype combinations yield positive population growth rates, especially if these effects are nonadditive, will be a critical component of an effective model.

Although rapid environmental change often involves selection on complex quantitative traits—for example, those involved in climate adaptation—the above theory has largely focused on fixation of small numbers of alleles affecting simple traits. In a brief review of recent studies examining climate-associated genomic variation (table A1), we show that climate adaptation is likely to be highly polygenic; genome-wide association study and genotype-environment association analyses found between 5 and 14,287 single nucleotide polymorphisms (SNPs) with signals for environmental selection, comprising 0.002%–4.6% of tested markers. For example, Bay and Palumbi (2014) scanned transcriptome-derived SNPs from reef-building corals in warmer and cooler habitats and identified 114 loci potentially under selection, showing allele frequency differences between habitats. In nearly half of the studies surveyed that did not examine quantitative trait loci (9/20), signals for selection covered more than 1% of markers, though only a single analysis estimated that greater than 3% of markers were under selection. Although we realize that these studies are not directly comparable—they vary in spatial scale, experimental and analytical methods, and significance cutoff—each study does represent a parameter estimate that could potentially be used in downstream prediction of climate adaptation. In addition, identification of loci under selection in natural populations suffers many pitfalls, including genomic coverage, demographic effects, and inability to accurately estimate phenotype and environment on relevant scales (reviewed in Hoban et al. 2016). Although estimation of genomic architecture of environmental tolerance is far from perfect, it is nonetheless clear that many of these traits are highly polygenic.

In cases where many loci contribute to environmental tolerance, changes in allele frequencies at many loci rather than fixation at a single locus may be sufficient to avert extinction (Latta 1998; Pritchard and Di Rienzo 2010). For this reason, efforts to predict the probability of adaptation to a changing environment in threatened populations will depend critically on determining the genomic architecture (the number of loci, their effects, and their distribution throughout the genome) of environmental tolerance in focal populations and translating distributions of many adaptive alleles into absolute fitness in different environments.

If environmental tolerance depends on many loci with high levels of standing variation, then quantitative trait models may be more relevant than population genetic models designed to capture fixation dynamics at a single locus (Pease et al. 1989; Lynch and Lande 1993; Gomulkiewicz and Holt 1995). However, quantitative genetic models often depend on estimates of heritable variation from common garden experiments or multigenerational field pedigrees, which are not feasible in many ecologically important species threatened by rapid environmental change, and they provide few tools for monitoring the genomic responses of populations to environmental change over time. The measurement of polygenic variation in environmental tolerance traits in natural populations remains a major challenge. When allelic effect estimates are available, patterns of polygenic variation can be measured by analyzing the distribution and abundance of alleles that affect a common trait (Berg and Coop 2014), but as mentioned above, the common gardens or pedigrees that enable such analyses are not available in many important systems.

Genotype-environment association approaches make the assumption that the distribution of alleles across space is a function of adaptation to contemporary environments, so each population is adapted to local conditions. A few studies of herbaceous plants use a combination of genotype-

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

environment association scans and common garden experiments to demonstrate that genetic variants associated with contemporary environmental variation can be used to predict performance under different environmental conditions (Hancock et al. 2011; Lasky et al. 2015). For example, genetic variants associated with environment in crop landraces of *Sorghum bicolor* predicted response (genotype by environment interactions [G × E]) to experimental drought and aluminum toxicity (Lasky et al. 2015). Additionally, climate-associated SNPs in *Arabidopsis thaliana* were predictive of relative fitness in a common garden (Hancock et al. 2011). Measurement of polygenic variation in wild populations of species that are not amenable to lab culture or common gardening is even more challenging. Genome-environment associations and analysis of allelic covariance may provide tools to address this problem (Wellenreuther and Hansson 2016), and some advancements have been made to predict polygenic scores without estimates of allelic effects (Bourret et al. 2014; Lasky et al. 2015). Further progress in the creation of multilocus models of genome-environment correlations will even further increase our ability to use genomic data to predict fitness in wild populations.

Although environmental tolerance traits may frequently be polygenic, the chromosomal distribution of adaptive variants will affect how selection acts on them (Tigano and Friesen 2016). If many adaptive variants are chromosomally linked—for instance, variants found on chromosomal inversions or in areas of low recombination—they may be less likely to be lost to genetic drift and more likely to contribute to rapid adaptation (Kirkpatrick and Barton 2006). In the extreme case, complex adaptations may be accomplished through the fixation of linked sets of adaptive variants, or supergenes, a case in which polygenic adaptation may be more accurately modeled with single-locus models (Schwander et al. 2014).

Spatial Distribution and Abundance of Adaptive Alleles

Predicting evolutionary responses to rapid environmental change is further complicated by environmental heterogeneity at multiple spatial and temporal scales, which in turn may alter the strength and response to selection across populations. A growing body of work studies the role that spatial heterogeneity plays in determining the rate and probability of rapid adaptation (Slatkin 1973; Ralph and Coop 2010; Uecker et al. 2014). Local adaptation to contemporary environmental extremes will in many cases maintain standing variation in environmental tolerance within species (Nagylaki 1975; Barton 1999; Polechová and Barton 2015). Yeaman (2015) recently investigated the possibility of local adaptation by alleles of small effect in the face of gene flow. They showed that even under extremely high gene flow, local adaptation can still evolve at the genome level if there is sufficient standing genetic variation, while no single locus makes an important contribution for long periods of time. For example, this might be the case in high gene flow marine systems: marine invertebrates inhabiting the pH gradient created by upwelling on the California coast exhibit genetic differentiation at a handful of genes despite a lack of overall structure caused by high dispersal (De Wit and Palumbi 2013; Pespeni et al. 2013). This maintenance of standing variation could contribute to adaptation to rapid environmental change.

If population fitness is reduced to below replacement by rapid environmental change, well-adapted individuals from previously extreme habitats may simply replace maladapted individuals as environmental change progresses. Under local dispersal, this will take roughly $d\sqrt{s}$ generations, where d is the distance the well-adapted alleles must travel in units of dispersal distance, and $1 + s$ is the growth rate of the well-adapted allele when rare (Fisher 1937). In addition, when gene flow brings environmental tolerance alleles to populations newly threatened by environmental change, these alleles can contribute to adaptation, a case sometimes referred to as genetic rescue (Tallmon et al. 2004; Dixon et al. 2015).

Gene flow, determined by habitat and population structure, therefore, impacts the likelihood of global or independent adaptation; when gene flow is low, populations will adapt more or less independently to the same novel environmental stress through convergent processes (Kaeuffer et al. 2012; Tenaillon et al. 2012; Roesti et al. 2014; Ralph and Coop 2015; Laporte et al. 2016; Oke et al. 2016). For example, a recent study by Laporte and colleagues (2016) combined a genetic map with a population genomic approach to assess whether parallel phenotypic changes in sympatric limnetic and benthic pairs of lake whitefish could be explained by genetic parallelism, multiple genetic routes, or both. The results revealed a highly polygenic architecture for body shape and provided evidence for both genetic parallelism and for multiple genetic routes underlying parallel phenotypic evolution in fish shape among populations occupying similar ecological niches. With sufficient gene flow, extreme populations can seed the process of evolutionary rescue across a species' range. Understanding the spatial scale across which a given adaptive process is expected to act is important for both defining the populations in which a given genomic architecture of adaptation is expected to apply and for defining units for management and monitoring of adaptation to environmental change (Schwartz et al. 2007). When populations are adapting to a patchily distributed novel environment, the transition between independent (i.e., convergent) versus shared adaptation happens at a critical distance between threatened patches, proportional to the probability of dispersal between threatened patches relative to the

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

selective disadvantage of an environmental tolerance allele between threatened patches (Slatkin 1973; Nagylaki 1975; Ralph and Coop 2015).

Phenotypic Plasticity

Phenotypic plasticity can play an important role in altering a population's response to rapid environmental change (Ghalambor et al. 2007; Lande 2009; Chevin et al. 2010, 2013). Plasticity can be adaptive, if it allows persistence under changed environmental conditions, or nonadaptive, if the resulting phenotype is further from the optimum (Ghalambor et al. 2007). Indeed, many of the studies documenting rapid phenotypic changes in response to environmental change reflect plasticity (Réale et al. 2003; Charmantier et al. 2008; Crozier et al. 2008; Hendry et al. 2008; Merilä and Hendry 2014). If populations can cope with rapid environmental changes via adaptive plasticity alone, it is unclear to what degree there is an opportunity for selection to result in an evolutionary response. Price et al. (2003) argue that evolutionary responses in the presence of adaptive plasticity should only occur if the phenotypic response lies below the local optimum, such that there is still an opportunity for selection. Indeed, in many cases where adaptive plasticity in gene expression has been suggested to facilitate adaptive evolution, the plastic changes appear to bring populations closer to the new optimum without reaching it (e.g., McCairns and Bernatchez 2010; Scoville and Pfrender 2010). Theoretical models of evolutionary rescue that incorporate adaptive phenotypic plasticity predict a decrease in the severity of a rapid environmental shift (i.e., a weakening of the strength of directional selection), which in turn slows the rate of population decline (Chevin et al. 2010).

Furthermore, if plasticity evolves but the phenotypic response is still below the optimum (Price et al. 2003), then the evolution of plasticity can increase the rate of adaptation (Lande 2009; Chevin et al. 2010). Alternatively, if plasticity is very costly, it may increase the probability of extinction (Chevin et al. 2010). More generally, such models confirm verbal arguments that adaptive plasticity can reduce extinction risk by increasing the range of environmental tolerance and helping populations persist in the face of environmental change (Chevin et al. 2010). However, novel environmental conditions can also induce nonadaptive or maladaptive plasticity, particularly when environment falls outside the range of historic conditions (Ghalambor et al. 2007). In such cases, nonadaptive plastic responses can potentially increase the risk of population extinction by shifting trait values further away from the new optimum and increasing the strength of directional selection. Thus, accounting for the degree to which traits respond to new environments through adaptive and or nonadaptive plastic responses will greatly increase the ability to make more accurate predictions on when selection can rescue populations. Transcriptomic studies using gene expression to examine the interactions and trade-offs between adaptation and plasticity show great promise for identifying and quantifying adaptive and nonadaptive plasticity (Palumbi et al. 2014; Ghalambor et al. 2015). These studies show that gene expression phenotypes under different environmental conditions can be used to quantify the strength of plasticity and population-level functional variation, even when the exact physiological mechanisms of adaptation are either unknown or difficult to measure.

Empirical Tests of Evolutionary Rescue

In recent years, there has been a proliferation of laboratory experiments testing theoretical predictions regarding evolutionary rescue in response to environmental change (Martin et al. 2013; Carlson et al. 2014). These experiments generally involve introducing populations to novel environments and allowing evolution to proceed through changes in frequency of existing genetic variants or the substitution of de novo mutations. Adaptation is typically quantified by comparing evolved populations to their ancestors. By introducing populations to sufficiently harsh environments that mean absolute fitness is initially reduced below one, researchers can test hypotheses about the various ecological, genetic, and demographic factors that can facilitate or impede evolutionary rescue.

One of the first explicit tests of the biological plausibility of evolutionary rescue theory was conducted by Bell and Gonzalez (2009), who exposed replicate populations of *Saccharomyces cerevisiae* to normally lethal concentrations of salt. They confirmed the general predictions of the theory, observing the expected U-shaped trajectory of population size over time in rescued populations and confirming the importance of initial population size for avoiding stochastic extinction while mean absolute fitness is below one. Other experiments with yeast and bacteria have similarly shown that large population sizes serve as a key buffer to help avoid reaching critical thresholds where extinction is likely (Bell and Gonzalez 2009, 2011; Samani and Bell 2010; Ramsayer et al. 2013). In addition to the importance of population size, experiments have confirmed theoretical predictions that the amount of standing genetic variation is a critical factor influencing the likelihood of evolutionary rescue. In the first explicit experimental test of this prediction, Agashe and colleagues exposed populations of flour beetles (*Tribolium castaneum*) to a novel corn flour resource and showed that rescue was positively correlated with initial levels of genetic variation (i.e., the number of founder laboratory strains; Agashe et al. 2011). Similar results have been obtained with bacterial populations subject to antibiotic stress

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

and unicellular green algae exposed to high levels of salt (Perron et al. 2008; Lachapelle and Bell 2012).

As discussed in the preceding section, additional key predictions of evolutionary rescue theory are that rescue will be more likely when the initial degree of maladaptation is small or environmental change is gradual. These predictions have been tested and supported in a number of experiments (Samani and Bell 2010; Bell and Gonzalez 2011; Gonzalez and Bell 2013; Lindsey et al. 2013). For instance, yeast populations with prior experience of exposure and adaptation to salt stress show greater likelihood of rescue than those that are naïve to this stressor (Gonzalez and Bell 2013). This experiment also supported theoretical predictions that gene flow can facilitate evolutionary rescue; metapopulations of yeast exposed to variable rates of environmental change across a spatial gradient of salinity stress were more likely to persist with local dispersal, provided that the rate of environmental change was not too great.

Knowledge Gaps and Limitations of Current Models

Detecting Polygenic Variation in Environmental Tolerance

As discussed above, most environmental tolerance traits appear to be under polygenic control. Thus, it is somewhat paradoxical that the vast majority of studies that investigate signatures of selection acting on wild plant or animal populations are based on genome scan methods that are biased toward the detection of large effects at single loci (Jensen et al. 2016). Several recent empirical studies have shown that alternative methods designed to detect signals of polygenic selection on the basis of allelic covariation among many small-effect loci have proved more efficient to detect the effect of selection (Berg and Coop 2014; Bourret et al. 2014; Pavey et al. 2015). However, if different populations have different genetic architectures of adaptation due to low levels of gene flow, insights derived from one population may have limited utility in others.

If environmental tolerance is so highly polygenic that determination of a useful genetic architecture explaining a large proportion of variation in tolerance is infeasible, an alternative approach to determining adaptive potential may be to quantify genetic variation in adaptive phenotypes under experimental conditions representative of future environments. Animal and plant breeders have long employed predictions of phenotypes (specifically, breeding values) based on genetic data, an approach known as marker-assisted selection (MAS) that now includes genomic selection (GS, using all markers; Lande and Thompson 1990; Heffner et al. 2009; Meuwissen and Goddard 2010). Briefly, these approaches work by first fitting a genotype to phenotype model in a set of training genotypes and then applying that model to predict phenotypes for additional genotypes. The accuracy of predictions from GS (Pearson's correlation r between predicted versus observed phenotypes) is proportional to the expected response from selection using these predictions (Heffner et al. 2009). A major challenge for using markers (as opposed to known causal variants) for phenotypic prediction is that predictions depend on linkage disequilibrium between markers and causal variants. As more recombination and selection events occur between the set of genotypes used to fit models and those being predicted, this linkage is likely to break down (due to recombination or fixation), necessitating recurring rounds of model fitting (Muir 2007). A combination of loci of both large effect and small effect may be important, as a result combining predictions from all markers, plus a subset of the most important markers may be most accurate (Bernardo 2014; Lasky et al. 2015; Spindel et al. 2015). If markers can be linked to absolute fitness, it might be possible to use predictions from MAS models to estimate future population persistence or decline.

Ecological Complexity of Selection

As biologists' ability to characterize molecular states has grown, their ability to measure and understand its ecological context has lagged. A central challenge in predicting evolutionary response to environmental change is that in most systems, researchers have a crude ability to tackle ecological complexity. For example, local adaptation along spatial gradients may be a poor proxy for response to future environments, if future environments do not closely match an occupied existing environment. It seems highly likely, given the high dimensionality of environmental variation (e.g., combined abiotic and biotic environments), that anthropogenic environmental change will tend to create no-analog selection regimes (i.e., that do correspond to any extant location; Williams and Jackson 2007; Fitzpatrick and Hargrove 2009).

Additionally, studies of local adaptation often use publicly available gridded estimates of climate and other variables for many populations across a species range (Hancock et al. 2011; Evans et al. 2014; Lasky et al. 2015), though these estimates can be crude and typically overlook the biotic environment. Even when selection can be measured in reciprocal transplants, it is rarely clear how applicable conditions in a given experiment are given high spatiotemporal environmental heterogeneity in nature (Ågren and Schemske 2012; Weigel and Nordborg 2015), especially for biotic conditions. Additional parameters critical to theory of evolutionary rescue, such as population growth rate, heritability, and dispersal ability are conditional on a given set of environmental conditions and can be notoriously challenging to estimate (Bjørnstad and Grenfell 2001; Clark et al. 2003).

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Integrative Models: The Response Architecture

Taken together, the theoretical and empirical advances have a major implication: efforts to understand the responses of populations to environmental change will depend on an understanding of not just the genetic architecture of adaptation but also the absolute fitness consequences, the spatial distribution and abundance of adaptive alleles, and the relationship of plasticity to adaptation in the system of interest. Motivated by this emerging consensus, we strongly advocate the increased study and modeling of evolutionary response architectures (fig. 1). This program will be challenging and touches on several major outstanding questions of population genetics; however, the challenge is urgent, and as reviewed above, a good deal of progress has been made.

Genetic Architecture

The ability to model or simulate the evolution of environmental tolerance traits will depend largely on good estimates of the genetic architecture of those traits. Ideally, a thorough understanding of genetic architecture should include the number of loci associated with environmental tolerance and their effect sizes, the locations of those loci within the genome, and the linkage structure among them. Our review of climate-related population genomic studies shows that measurements of these parameters in natural populations are highly variable (table A1). Important general principles include that climate-associated traits are likely to be polygenic and distributed across a broad range of cellular functions. This may be an unsatisfying result if the goal is one smoking gun gene, characteristic of early evolutionary models. However, modern population genomic approaches for study of complex traits may lead us to focus on generalizable patterns rather than searching for a single gene of known function. As methods for identifying and interpreting the genomic basis of highly polygenic traits improve, so will our ability to use these inferences to predict evolutionary outcomes.

Spatial Distribution and Abundance of Adaptive Alleles

Although great progress is being made in characterizing the genomic architecture of environmental adaptation, this is just part of the challenge of understanding evolutionary responses to environmental change. We next need estimates of the spatial distribution and abundance of putatively adaptive alleles. This should consist of allele frequencies at all associated loci across a region of interest in order to predict the impacts of standing variation and migration on the probability of adaptation. Although for highly polygenic traits, procuring this type of data can be costly, decreases in sequencing costs and advances in genotyping from low-coverage genomes are allowing increased sample sizes and broader sampling. In addition, informative polymorphisms inferred from genomic data can be used to develop smaller, more cost-effective assays (Ruegg et al. 2014; Jones and Good 2016). Understanding the distribution of standing genetic variation will allow for population-specific simulations of future spread of adaptive genotypes.

Population Dynamics

Connecting our understanding of the genomic basis of adaptation to predictions of population persistence will require linking inferred genetic architecture of environmental tolerance with accurate estimates of fitness under different environmental conditions, both biotic and abiotic. A major gap in knowledge is the disconnect between relative fitness, which has been incorporated into the framework of population genetics for decades, and absolute fitness, which is necessary for modeling changes in population size during adaptation to novel environmental stress (Lynch and Lande 1993; Orr 2009). The best estimates are gathered from direct measures of fecundity under known conditions, but this data is not possible for all organisms. Combining measures of relative fitness inferred from genome-environment association studies with population growth rates measured during colonization of new or disturbed areas might provide us with rough estimates of absolute fitness for species that are difficult to keep in the lab or for which fecundity is not easy to measure. The development of standard methods for integrating these types of data will be an important next step toward using population genomic data to predict changes in future population size.

Plasticity

Finally, prediction of response to environmental change will necessarily include knowledge about the relative roles of adaptation and plasticity as well as interactions between these two processes. Much has been learned about the magnitude of plasticity and potential trade-offs from transplant studies in natural systems, aided by the use transcriptome data to investigate gene expression phenotypes (Palumbi et al. 2014; Ghalambor et al. 2015; Rose et al. 2016). In order to incorporate plasticity into predictions of future response, we must be able to measure how flexible environmental tolerance traits are and the effect of plasticity on the rate of adaptation. Although this is still a major outstanding question in evolutionary biology, focused experimentation in natural systems as well as in the laboratory may help measure these parameters in specific systems for incorporation into models of environmental response.

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Future Directions

Some studies have already attempted to address multiple elements of this proposed response architecture in determining population responses to evolutionary change (Chevin et al. 2010; Fournier-Level et al. 2016), but very few have simultaneously measured or modeled these disparate influences on population persistence. Finally we urge increased comparison across studies and across systems. Although new genomic studies are conducted every month, results have rarely been used to search for general principles of rapid adaptation or to prioritize conservation efforts (but see Aitken et al. 2008; Shafer et al. 2015). Meta-analyses and broadly comparative studies could be highly informative for these questions (e.g., Thurman and Barrett 2016). This requires increased transparency and sharing of data but could result in findings of extremely high utility. Combining results across multiple studies will increase our estimation accuracy for mechanisms of environmental tolerance and therefore our ability to predict the fate of species in a rapidly changing world.

Acknowledgments

The ideas for this article originated through a symposium at the American Society of Naturalists' annual meeting. For this reason, we gratefully acknowledge the society as well as the organizers of that meeting.

APPENDIX

Supplemental Table

Table A1: Studies from 2014–2015 examining the genomic basis for adaptation to climate

Study	Environment/phenotype	Design	Scale (km)	Total loci	Climate loci
Hordeum vulgare (Abebe et al. 2015)	Altitude, rainfall, temperature	G-E	40	1,370	40
Canis lupis (Stronen et al. 2015)	12 climate variables	G-E	2,700	67,000	175
Parus humilis (Qu et al. 2015)	Altitude	G-E	3,500	3,613,365	183
Homo sapiens (Fumagalli et al. 2015)	Cardiometabolic phenotype	GWAS	8,400	76,872	20
Iris hexagona (Hamlin and Arnold 2015)	19 climate variables	G-E	1,000	750	35
Corallium rubrum (Pratlong et al. 2015)	Depth	G-E	.6	237,681	56
Arabidopsis thaliana (Raschke et al. 2015)	Temperature-associated growth	QTL	…	…	1
Zea mays (Takuno et al. 2015)	Altitude	G-E	200	76,989	687
Sitta carolinensis (Manthey and Moyle 2015)	19 climate variables	G-E	412	2,635	43
Oryza glaberrima (Li et al. 2015)	Thermotolerance	QTL	…	…	5
Oryza rufipogon (Mao et al. 2015)	Low-temperature performance	QTL	…	…	13
Gadus morhua (Berg et al. 2015)	Salinity, temperature, oxygen	G-E	800	8,809	234
Gasterosteus aculeatus (Guo et al. 2015)	Salinity, temperature	G-E	1,000	9,404	152
O. rufipogon (Ma et al. 2015)	Cold tolerance	QTL	…	…	5
A. thaliana (Lobréaux and Melodelima 2015)	Temperature	G-E	7,700	948,330	14,287
Drosophila serrata (Latimer et al. 2015)	Thermal performance curve	QTL	…	…	11
Acropora hyacinthus (Bay and Palumbi 2014)	Temperature	G-E	1	15,399	114
Cryptomeria japonica (Tsumura et al. 2014)	19 climate variables	G-E	900	3,930	38
Ovis aries (Lv et al. 2014)	105 climate variables	G-E	11,000	49,034	230
Oncorhynchus gorbuscha (Limborg et al. 2014)	Latitude	G-E	2,800	8,036	101
Alnus glutinosa (De Kort et al. 2014)	8 climate variables	G-E	1,500	1,990	15
Populus trichocarpa (Evans et al. 2014)	20 climate variables	G-E	1,100	3,940,000	397
A. thaliana (Oakley et al. 2014)	Freezing tolerance	QTL	…	…	7
Restio capensis (Lexer et al. 2014)	19 climate variables	G-E	500	14,434	156
Brassica napus (Nelson et al. 2014)	Thermal time to flowering	QTL	…	…	3
Eucalyptus tricarpa (Steane et al. 2014)	Aridity	G-E	250	6,544	94
Oncorhynchus tshawytscha (Everett and Seeb 2014)	Thermal tolerance	QTL	…	…	3
Sorghum bicolor (Bekele et al. 2014)	Germination and growth under cold stress	QTL	…	…	61
A. thaliana (Li et al. 2014)	Thermal sensitivity	GWAS	…	213,000	5
Rubus idaeus (Molina-Bravo et al. 2014)	Chilling requirement	QTL	…	…	4

Note: Studies are listed in reverse chronological order, with newest studies first. G-E = genotype-environment association; GWAS = genome-wide association study; QTL = quantitative trait locus mapping.

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Literature Cited

Agashe, D., J. J. Falk, and D. I. Bolnick. 2011. Effects of founding genetic variation on adaptation to a novel resource. Evolution 65: 2481–2491.

Ågren, J., and D. W. Schemske. 2012. Reciprocal transplants demonstrate strong adaptive differentiation of the model organism *Arabidopsis thaliana* in its native range. New Phytologist 194:1112–1122.

Aitken, S. N., S. Yeaman, J. A. Holliday, T. Wang, and S. Curtis-McLane. 2008. Adaptation, migration or extirpation: climate change outcomes for tree populations. Evolutionary Applications 1:95–111.

Balanyá, J., J. M. Oller, R. B. Huey, G. W. Gilchrist, and L. Serra. 2006. Global genetic change tracks global climate warming in *Drosophila subobscura*. Science 313:1773–1775.

Barrett, R. D. H., and D. Schluter. 2008. Adaptation from standing genetic variation. Trends in Ecology and Evolution 23:38–44.

Barton, N. H. 1999. Clines in polygenic traits. Genetical Research 74: 223–236.

Bay, R. A., and S. R. Palumbi. 2014. Multilocus adaptation associated with heat resistance in reef-building corals. Current Biology 24:2952–2956.

Bell, G., and A. Gonzalez. 2009. Evolutionary rescue can prevent extinction following environmental change. Ecology Letters 12:942–948.

———. 2011. Adaptation and evolutionary rescue in metapopulations experiencing environmental deterioration. Science 332:1327–1330.

Berg, J. J., and G. Coop. 2014. A population genetic signal of polygenic adaptation. PLoS Genetics 10:e1004412.

Bernardo, R. 2014. Genomewide selection when major genes are known. Crop Science 54:68–75.

Bjørnstad, O. N., and B. T. Grenfell. 2001. Noisy clockwork: time series analysis of population fluctuations in animals. Science 293:638–643.

Bourret, V., M. Dionne, and L. Bernatchez. 2014. Detecting genotypic changes associated with selective mortality at sea in Atlantic salmon: polygenic multilocus analysis surpasses genome scan. Molecular Ecology 23:4444–4457.

Bradshaw, W. E., and C. M. Holzapfel. 2001. Genetic shift in photoperiodic response correlated with global warming. Proceedings of the National Academy of Sciences of the USA 98:14509–14511.

Burger, R., and M. Lynch. 1995. Evolution and extinction in a changing environment: a quantitative-genetic analysis. Evolution 49:151–163.

Carlson, S. M., C. J. Cunningham, and P. A. H. Westley. 2014. Evolutionary rescue in a changing world. Trends in Ecology and Evolution 29:521–530.

Charmantier, A., R. H. McCleery, L. R. Cole, C. Perrins, L. E. B. Kruuk, and B. C. Sheldon. 2008. Adaptive phenotypic plasticity in response to climate change in a wild bird population. Science 320:800–803.

Chevin, L.-M., R. Gallet, R. Gomulkiewicz, R. D. Holt, and S. Fellous. 2013. Phenotypic plasticity in evolutionary rescue experiments. Philosophical Transactions of the Royal Society B 368:20120089.

Chevin, L.-M., R. Lande, and G. M. Mace. 2010. Adaptation, plasticity, and extinction in a changing environment: towards a predictive theory. PLoS Biology 8:e1000357.

Clark, J. S., M. Lewis, J. S. McLachlan, and J. HilleRisLambers. 2003. Estimating population spread: what can we forecast and how well? Ecology 84:1979–1988.

Collins, S., and G. Bell. 2004. Phenotypic consequences of 1,000 generations of selection at elevated CO_2 in a green alga. Nature 431:566–569.

Crow, J. F. 1970. Genetic loads and the cost of natural selection. Pages 128–177 *in* K. Kojima, ed. Mathematical topics in population genetics: biomathematics. Springer, Berlin.

Crozier, L. G., A. P. Hendry, P. W. Lawson, T. P. Quinn, N. J. Mantua, J. Battin, R. G. Shaw, and R. B. Huey. 2008. Potential responses to climate change in organisms with complex life histories: evolution and plasticity in Pacific salmon. Evolutionary Applications 1:252–270.

De Wit, P., and S. R. Palumbi. 2013. Transcriptome-wide polymorphisms of red abalone (*Haliotis rufescens*) reveal patterns of gene flow and local adaptation. Molecular Ecology 22:2884–2897.

Dixon, G. B., S. W. Davies, G. A. Aglyamova, E. Meyer, L. K. Bay, and M. V. Matz. 2015. Genomic determinants of coral heat tolerance across latitudes. Science 348:1460–1462.

Evans, L. M., G. T. Slavov, E. Rodgers-Melnick, J. Martin, P. Ranjan, W. Muchero, A. M. Brunner, et al. 2014. Population genomics of *Populus trichocarpa* identifies signatures of selection and adaptive trait associations. Nature Genetics 46:1089–1096.

Fernandes, J. A., W. W. L. Cheung, S. Jennings, M. Butenschön, L. de Mora, T. L. Frölicher, M. Barange, and A. Grant. 2013. Modelling the effects of climate change on the distribution and production of marine fishes: accounting for trophic interactions in a dynamic bioclimate envelope model. Global Change Biology 19:2596–2607.

Fisher, R. A. 1937. The wave of advance of advantageous genes. Annals of Eugenics 7:355–369.

Fitzpatrick, M. C., and W. W. Hargrove. 2009. The projection of species distribution models and the problem of non-analog climate. Biodiversity and Conservation 18:2255–2261.

Fournier-Level, A., E. O. Perry, J. A. Wang, P. T. Braun, A. Migneault, M. D. Cooper, C. J. E. Metcalf, and J. Schmitt. 2016. Predicting the evolutionary dynamics of seasonal adaptation to novel climates in *Arabidopsis thaliana*. Proceedings of the National Academy of Sciences of the USA 201517456.

Ghalambor, C. K., K. L. Hoke, E. W. Ruell, E. K. Fischer, D. N. Reznick, and K. A. Hughes. 2015. Non-adaptive plasticity potentiates rapid adaptive evolution of gene expression in nature. Nature 525:372–375.

Ghalambor, C. K., J. K. McKay, S. P. Carroll, and D. N. Reznick. 2007. Adaptive versus non-adaptive phenotypic plasticity and the potential for contemporary adaptation in new environments. Functional Ecology 21:394–407.

Gomulkiewicz, R., and R. D. Holt. 1995. When does evolution by natural selection prevent extinction? Evolution 49:201–207.

Gonzalez, A., and G. Bell. 2013. Evolutionary rescue and adaptation to abrupt environmental change depends upon the history of stress. Philosophical Transactions of the Royal Society B 368:20120079.

Hancock, A. M., B. Brachi, N. Faure, M. W. Horton, L. B. Jarymowycz, F. G. Sperone, C. Toomajian, F. Roux, and J. Bergelson. 2011. Adaptation to climate across the *Arabidopsis thaliana* genome. Science 334:83–86.

Heffner, E. L., M. E. Sorrells, and J.-L. Jannink. 2009. Genomic selection for crop improvement. Crop Science 49:1–12.

Hendry, A. P., T. J. Farrugia, and M. T. Kinnison. 2008. Human influences on rates of phenotypic change in wild animal populations. Molecular Ecology 17:20–29.

Hendry, A. P., and M. T. Kinnison. 1999. Perspective: the pace of modern life—measuring rates of contemporary microevolution. Evolution 53:1637–1653.

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Hermisson, J., and P. S. Pennings. 2005. Soft sweeps molecular population genetics of adaptation from standing genetic variation. Genetics 169:2335–2352.

Hoban, S., J. L. Kelley, K. E. Lotterhos, M. F. Antolin, G. Bradburd, D. B. Lowry, M. L. Poss, L. K. Reed, A. Storfer, and M. C. Whitlock. 2016. Finding the genomic basis of local adaptation: pitfalls, practical solutions, and future directions. American Naturalist 188:379–397.

Hoffmann, A. A., and C. M. Sgrò. 2011. Climate change and evolutionary adaptation. Nature 470:479–485.

Holt, R. D., M. Barfield, R. Gomulkiewicz, and A. E. P. Lundberg. 2004. Temporal variation can facilitate niche evolution in harsh sink environments. American Naturalist 164:187–200.

Jensen, J. D., M. Foll, and L. Bernatchez. 2016. The past, present and future of genomic scans for selection. Molecular Ecology 25:1–4.

Jones, M. R., and J. M. Good. 2016. Targeted capture in evolutionary and ecological genomics. Molecular Ecology 25:185–202.

Kaeuffer, R., C. L. Peichel, D. I. Bolnick, and A. P. Hendry. 2012. Parallel and nonparallel aspects of ecological, phenotypic, and genetic divergence across replicate population pairs of lake and stream stickleback. Evolution 66:402–418.

Kirkpatrick, M., and N. Barton. 2006. Chromosome inversions, local adaptation and speciation. Genetics 173:419–434.

Lachapelle, J., and G. Bell. 2012. Evolutionary rescue of sexual and asexual populations in a deteriorating environment. Evolution 66: 3508–3518.

Lande, R. 2009. Adaptation to an extraordinary environment by evolution of phenotypic plasticity and genetic assimilation. Journal of Evolutionary Biology 22:1435–1446.

Lande, R., and R. Thompson. 1990. Efficiency of marker-assisted selection in the improvement of quantitative traits. Genetics 124:743–756.

Laporte, M., S. A. Pavey, C. Rougeux, F. Pierron, M. Lauzent, H. Budzinski, P. Labadie, et al. 2016. RAD sequencing reveals within-generation polygenic selection in response to anthropogenic organic and metal contamination in North Atlantic eels. Molecular Ecology 25:219–237.

Lasky, J. R., H. D. Upadhyaya, P. Ramu, S. Deshpande, C. T. Hash, J. Bonnette, T. E. Juenger, et al. 2015. Genome-environment associations in sorghum landraces predict adaptive traits. Science Advances 1:e1400218.

Latta, R. G. 1998. Differentiation of allelic frequencies at quantitative trait loci affecting locally adaptive traits. American Naturalist 151: 283–292.

Lindsey, H. A., J. Gallie, S. Taylor, and B. Kerr. 2013. Evolutionary rescue from extinction is contingent on a lower rate of environmental change. Nature 494:463–467.

Lynch, M., and R. Lande. 1993. Evolution and extinction in response to environmental change. Pages 234-250 *in* P. Kareiva, J. Kingsolver, and R. Huey, eds. Biotic interactions and global change. Sinauer, Sunderland, MA.

Martin, G., R. Aguilée, J. Ramsayer, O. Kaltz, and O. Ronce. 2013. The probability of evolutionary rescue: towards a quantitative comparison between theory and evolution experiments. Philosophical Transactions of the Royal Society B 368:20120088.

McCairns, R. J., and L. Bernatchez. 2010. Adaptive divergence between freshwater and marine sticklebacks: insights into the role of phenotypic plasticity from an integrated analysis of candidate gene expression. Evolution 64:1029–1047.

Merilä, J., and A. P. Hendry. 2014. Climate change, adaptation, and phenotypic plasticity: the problem and the evidence. Evolutionary Applications 7:1–14.

Messer, P. W., and D. A. Petrov. 2013. Population genomics of rapid adaptation by soft selective sweeps. Trends in Ecology and Evolution 28:659–669.

Meuwissen, T., and M. Goddard. 2010. Accurate prediction of genetic values for complex traits by whole-genome resequencing. Genetics 185:623–631.

Muir, W. M. 2007. Comparison of genomic and traditional BLUP-estimated breeding value accuracy and selection response under alternative trait and genomic parameters. Journal of Animal Breeding and Genetics 124:342–355.

Nagylaki, T. 1975. Conditions for the existence of clines. Genetics 80:595–615.

Oke, K. B., M. Bukhari, R. Kaeuffer, G. Rolshausen, K. Räsänen, D. I. Bolnick, C. L. Peichel, and A. P. Hendry. 2016. Does plasticity enhance or dampen phenotypic parallelism? a test with three lake-stream stickleback pairs. Journal of Evolutionary Biology 29:126–143.

Orr, H. A. 2009. Fitness and its role in evolutionary genetics. Nature Reviews Genetics 10:531–539.

Orr, H. A., and R. L. Unckless. 2014. The population genetics of evolutionary rescue. PLoS Genetics 10:e1004551.

Orr, H. A., R. L. Unckless, A. E. Whitlock, and E. M. C. Whitlock. 2008. Population extinction and the genetics of adaptation. American Naturalist 172:160–169.

Palumbi, S. R., D. J. Barshis, N. Traylor-Knowles, and R. A. Bay. 2014. Mechanisms of reef coral resistance to future climate change. Science 344:895–898.

Paul, J. R., S. N. Sheth, A. L. Angert, and S. E. M. A. Geber. 2011. Quantifying the impact of gene flow on phenotype-environment mismatch: a demonstration with the scarlet monkeyflower *Mimulus cardinalis*. American Naturalist 178(suppl.):S62–S79.

Pavey, S. A., J. Gaudin, E. Normandeau, M. Dionne, M. Castonguay, C. Audet, and L. Bernatchez. 2015. RAD sequencing highlights polygenic discrimination of habitat ecotypes in the panmictic American eel. Current Biology 25:1666–1671.

Pease, C. M., R. Lande, and J. J. Bull. 1989. A model of population growth, dispersal and evolution in a changing environment. Ecology 70:1657–1664.

Perron, G. G., A. Gonzalez, and A. Buckling. 2008. The rate of environmental change drives adaptation to an antibiotic sink. Journal of Evolutionary Biology 21:1724–1731.

Pespeni, M. H., E. Sanford, B. Gaylord, T. M. Hill, J. D. Hosfelt, H. K. Jaris, M. LaVigne, et al. 2013. Evolutionary change during experimental ocean acidification. Proceedings of the National Academy of Sciences of the USA 110:6937–6942.

Pinsky, M. L., B. Worm, M. J. Fogarty, J. L. Sarmiento, and S. A. Levin. 2013. Marine taxa track local climate velocities. Science 341:1239–1242.

Polechová, J., and N. H. Barton. 2015. Limits to adaptation along environmental gradients. Proceedings of the National Academy of Sciences of the USA 112:6401–6406.

Price, T. D., A. Qvarnström, and D. E. Irwin. 2003. The role of phenotypic plasticity in driving genetic evolution. Proceedings of the Royal Society of London B 270:1433–1440.

Pritchard, J. K., and A. Di Rienzo. 2010. Adaptation—not by sweeps alone. Nature Reviews Genetics 11:665–667.

Ralph, P., and G. Coop. 2010. Parallel adaptation: one or many waves of advance of an advantageous allele? Genetics 186:647–668.

Ralph, P. L., and G. Coop. 2015. Convergent evolution during local adaptation to patchy landscapes. PLoS Genetics 11:e1005630.

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).

Ramsayer, J., O. Kaltz, and M. E. Hochberg. 2013. Evolutionary rescue in populations of *Pseudomonas fluorescens* across an antibiotic gradient. Evolutionary Applications 6:608–616.

Réale, D., A. G. McAdam, S. Boutin, and D. Berteaux. 2003. Genetic and plastic responses of a northern mammal to climate change. Proceedings of the Royal Society of London B 270:591–596.

Reznick, D. N., and C. K. Ghalambor. 2001. The population ecology of contemporary adaptations: what empirical studies reveal about the conditions that promote adaptive evolution. Genetica 112–113: 183–198.

Roesti, M., S. Gavrilets, A. P. Hendry, W. Salzburger, and D. Berner. 2014. The genomic signature of parallel adaptation from shared genetic variation. Molecular Ecology 23:3944–3956.

Rose, N. H., F. O. Seneca, and S. R. Palumbi. 2016. Gene networks in the wild: identifying transcriptional modules that mediate coral resistance to experimental heat stress. Genome Biology and Evolution 8:243–252.

Ruegg, K., E. C. Anderson, J. Boone, J. Pouls, and T. B. Smith. 2014. A role for migration-linked genes and genomic islands in divergence of a songbird. Molecular Ecology 23:4757–4769.

Samani, P., and G. Bell. 2010. Adaptation of experimental yeast populations to stressful conditions in relation to population size. Journal of Evolutionary Biology 23:791–796.

Schwander, T., R. Libbrecht, and L. Keller. 2014. Supergenes and complex phenotypes. Current Biology 24:R288–R294.

Schwartz, M. K., G. Luikart, and R. S. Waples. 2007. Genetic monitoring as a promising tool for conservation and management. Trends in Ecology and Evolution 22:25–33.

Scoville, A. G., and M. E. Pfrender. 2010. Phenotypic plasticity facilitates recurrent rapid adaptation to introduced predators. Proceedings of the National Academy of Sciences of the USA 107:4260–4263.

Shafer, A. B. A., J. B. W. Wolf, P. C. Alves, L. Bergström, M. W. Bruford, I. Brännström, G. Colling, et al. 2015. Genomics and the challenging translation into conservation practice. Trends in Ecology and Evolution 30:78–87.

Slatkin, M. 1973. Gene flow and selection in a cline. Genetics 75:733–756.

Somero, G. N. 2010. The physiology of climate change: how potentials for acclimatization and genetic adaptation will determine "winners" and "losers." Journal of Experimental Biology 213:912–920.

Spindel, J., H. Begum, D. Akdemir, P. Virk, B. Collard, E. Redoña, G. Atlin, J.-L. Jannink, and S. R. McCouch. 2015. Genomic selection and association mapping in rice (*Oryza sativa*): effect of trait genetic architecture, training population composition, marker number and statistical model on accuracy of rice genomic selection in elite, tropical rice breeding lines. PLoS Genetics 11:e1004982.

Tallmon, D. A., G. Luikart, and R. S. Waples. 2004. The alluring simplicity and complex reality of genetic rescue. Trends in Ecology and Evolution 19:489–496.

Tenaillon, O., A. Rodríguez-Verdugo, R. L. Gaut, P. McDonald, A. F. Bennett, A. D. Long, and B. S. Gaut. 2012. The molecular diversity of adaptive convergence. Science 335:457–461.

Thurman, T. J., and R. D. Barrett. 2016. The genetic consequences of selection in natural populations. Molecular Ecology 25:1429–1448.

Tigano, A., and V. L. Friesen. 2016. Genomics of local adaptation with gene flow. Molecular Ecology 25:2144–2164.

Uecker, H., S. P. Otto, and J. Hermisson. 2014. Evolutionary rescue in structured populations. American Naturalist 183:E17–E35.

Umina, P. A., A. R. Weeks, M. R. Kearney, S. W. McKechnie, and A. A. Hoffmann. 2005. A rapid shift in a classic clinal pattern in *Drosophila* reflecting climate change. Science 308:691–693.

Van Doorslaer, W., R. Stoks, E. Jeppesen, and L. De Meester. 2007. Adaptive microevolutionary responses to simulated global warming in *Simocephalus vetulus*: a mesocosm study. Global Change Biology 13:878–886.

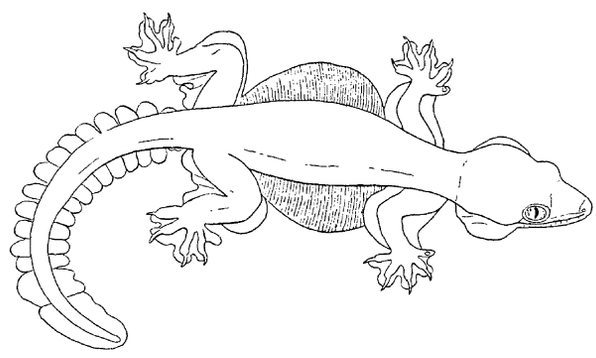
Weigel, D., and M. Nordborg. 2015. Population genomics for understanding adaptation in wild plant species. Annual Review of Genetics 49:315–338.

Wellenreuther, M., and B. Hansson. 2016. Detecting polygenic evolution: problems, pitfalls, and promises. Trends in Genetics 32:155–164.

Williams, J. W., and S. T. Jackson. 2007. Novel climates, no-analog communities, and ecological surprises. Frontiers in Ecology and the Environment 5:475–482.

Yeaman, S. 2015. Local adaptation by alleles of small effect. American Naturalist 186(suppl.):S74–S89.

Associate Editor: Scott L. Nuismer
Editor: Yannis Michalakis



"Among modern types two lizards have the power of soaring, though in varying degree. Of these lizards the first is of the family Geckonidæ, *Ptychozoön homalocephalum*, a remarkable form from the Malay Islands and the Malay Peninsula." From "Volant Adaptation in Vertebrates" by Richard S. Lull (*The American Naturalist*, 1906, 40:537–566).

This content downloaded from 128.097.027.020 on May 30, 2017 14:46:17 PM
All use subject to University of Chicago Press Terms and Conditions (http://www.journals.uchicago.edu/t-and-c).